



MASTER'S DEGREE PROJECT IN MATHEMATICS

**On Algebras Induced by Novel Involutions in the Cayley–Dickson
Construction**

by

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Abstract

The Cayley–Dickson construction generates new algebras of twice the dimension of the standard algebra. The construction is commonly done by starting with the field of real numbers, $\mathbb{R}$. By utilizing the standard involution, one obtains the complex numbers, the quaternions and the octonions with the standard choice of parameter of $\mu = -1$. The real numbers, the complex numbers, the quaternions and the octonions can be shown to be the only real normed division algebras. This thesis will focus on explaining the Cayley–Dickson construction by walking through the construction of these four normed division algebras. Furthermore, we will show that a different mapping is an involution and examine what algebras are obtained through the Cayley–Dickson construction with that choice of mapping.

Keywords: Cayley–Dickson construction, Complex numbers, Quaternions, Octonions, Involution

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Chapter 1

History and Introduction

The Cayley–Dickson construction is an interesting process to generate new algebras from previous algebras as in each step of the process the dimensionality of the algebra doubles. Ultimately, the price one pays is the loss of some important properties such as associativity, commutativity, and so on [5]. Furthermore, the Cayley–Dickson may extend indefinitely, although there are only four real normed division algebras [4][5][16][21]. Stranger is the fact that the quaternions were discovered in 1843 by Hamilton [5][12]. This discovery was made long before the rigorous definition of the Cayley–Dickson construction itself [4][5]. Hamilton had found a way to construct complex numbers as pairs of real numbers at the age of 30, Hamilton made this discovery in 1835 [5]. Hamilton had continued his idea of extending a pair of real numbers to a triple of real numbers. The process for Hamilton was time consuming and many failed attempts were made before finally reaching the quaternions in 1843 [5][12]. Hamilton made the discovery whilst walking with his wife down the Royal Canal in Dublin on October 16, then Hamilton realized that he needed yet another *imaginary* symbol for the process to work [5][12]. Moreover, Hamilton could not resist the impulse to carve out the right relationship between the imaginary symbols in a stone on the Brougham Bridge [5][12]. As Baez discloses in his paper *The octonions*[5]

“ That is to say, I then and there felt the galvanic circuit of thought close; and the sparks which fell from it were the fundamental equations between i, j, k ; exactly such as I have used them ever since.”

Although many strictly skewed fields have been discovered since Hamilton’s days, the quaternions were the first known strictly skewed field [12]. Ultimately in 1909 Wedderburn proved that none of the strictly skewed fields were finite [12]. Likewise, Wedderburn proved a similar result for rational division algebras of order eight [1].

The octonions were discovered around the same time by Hamilton’s old friend, Graves [5]. In fact, it was Graves who inspired Hamilton to think about real numbers as pairs or triplets. In October of 1843 Hamilton sent his results to Graves, and Graves remarked the following in a reply to Hamilton [5]

“ . . . If with your alchemy you can make three pounds of gold, why should you stop there?”

By setting his aim on even more imaginary symbols, Graves made the discovery of what he

would call *the octaves*¹ [5]. Graves readdressed this to Hamilton in December of 1843 where he concluded that the octonions were a normed division algebra and could be used to express the product of two sums of eight perfect squares as another sum of eight perfect squares, the so-called *eight squares theorem* [5][11]. It is not known why Graves did not choose to publish his discovery at the time, although Hamilton urged and offered to help him with the publication [5]. Instead, Graves continued his work on algebras of 2^n dimensions and called them just that 2^n -ions, but the work was unfruitful as he began to doubt whether more existed or not [5]. Hamilton continued his responses with Graves and coined the term *associative* when he realized that the octonions did not satisfy $a(bc) = (ab)c$.

Cayley was fascinated by belongings and is said to be the first to give an abstract definition of a group [12]. Later, Cayley started the work of what would become the now known Cayley–Dickson construction [4][10]. Cayley had been fascinated by the work of Hamilton and the existence of quaternions. Cayley sought a relation between hyperelliptic functions and the quaternions and published his results in 1845 in *Philosophical Magazine* with the title “On Jacobi’s Elliptic Functions, in Reply to Rev. B. Bronwin; and on Quaternions” [5][10]. Although strange, Cayley tackled the octonions in an appendix in the paper and the work on hyperelliptic functions was full of errors so that everything but the octonions were retracted from Cayley’s collected works [5][9]. Graves did not take the publication lightly and tried to argue that he had known of the existence of the octonions since 1843 and Hamilton even tried to vouch for Graves’ sake, with no effect. The octonions came to be known as *Cayley numbers* and Graves were effectively shattered, even worse is that the eight-squares theorem had already been discovered earlier by Degen in 1818 [5][11].

Dickson continued Cayley’s work on octonions and Cayley numbers to what would eventually become the *Cayley–Dickson construction* and therefore the construction is named after both Arthur Cayley and Leonard Dickson [4][11]. During the 20th century, Albert generalized the Cayley–Dickson construction through a more arbitrary product by introducing a parameter in the product [2][4]. Similarly, Schafer [19] proved that beyond octonions, algebras are no longer composition algebras [16]. Urbanik and Wright [21] proved that every arbitrary absolute-valued algebra is isomorphic to the real numbers, the complex numbers, the quaternions or the octonions, building on the work of Albert[3]. Brown’s work on Generalized Cayley–Dickson algebras was published in 1967 and concerns with classification of algebras of dimensions 16, 32 and 64 and when the 16-dimensional algebras are division algebras was likewise formulated by Brown [8]. In 1985 McCrimmon proved Schafer’s results by not assuming the characteristic of the underlying field and making use of the Cayley derivation [16]. Ultimately, the Cayley–Dickson construction in its standard form may have eight different products, as Bales [6][7] describes. Rarely have more than two of these products been researched [7]. Likewise, in its generalized form, as discovered by Albert, there should be even more products. The Cayley–Dickson algebras form a twisted group, but little to no interest has been paid for this twisted group [6].

¹Hereafter referred to as the name used today: octonions

In this thesis, the Cayley–Dickson construction will be explained, what the construction is and what kind of algebras the construction generates when starting with the algebra $\mathbb{R}$ as seen over $\mathbb{R}$. The first four algebras through the standard involution and standard product on the algebras will be explained, as discovered by Cayley and Dickson, although we will also comment on the more general product introduced by Albert. Furthermore, we will explain why these four algebras are the only four real normed division algebras and explain the norm on these algebras. Thereafter, we will examine and explain whether these algebras are also inner product spaces. Furthermore, we will observe another natural mapping and show that this mapping is an involution on the Cayley–Dickson construction. What kind of algebras are generated by this involution will likewise be examined.

Chapter 2

Preliminaries

2.1 Abstract Algebra

This section introduces the necessary theory about abstract algebra. First, recall the following definition of a *group* [12].

Definition 1 (Group). A *group* is a set G , which is closed under a binary operation, $*$ and denoted by G , such that the following are satisfied:

$\mathcal{G}_1$: $\forall g, h, i \in G$, it follows that

$$g * (h * i) = (g * h) * i.$$

We call this the *associative* property for G under the operation $*$.

$\mathcal{G}_2$: $\exists 0_G \in G$ such that $\forall g \in G$ the following is true

$$0_G * g = g * 0_G = g.$$

We refer to 0_G as the *identity element* for the operation $*$ in the group G .

$\mathcal{G}_3$: For each $g \in G$ there is an element $g^{-1} \in G$ such that

$$g * g^{-1} = g^{-1} * g = 0_G.$$

We call g^{-1} the *inverse* of g .

Definition 2 (Abelian group). Let G be a group. Then, if the following property is also satisfied, the group is called an *abelian group*.

$\mathcal{G}_4$: $\forall g, h \in G$, it holds that

$$g * h = h * g.$$

We call the property in $\mathcal{G}_4$ the *commutative* property of the group G under the operation $*$.

For abelian groups, we denote the binary operation $*$ by the ordinary plus sign $+$.

Example 1. Common abelian groups are $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ under the operation of ordinary addition.

Groups are algebraic structures with one binary operation. If we impose yet another binary operation on a group, we obtain a more complex structure.

Definition 3 (Ring). A *ring* is a set R with two binary operations $*$ and $\cdot$. Both operations are closed in the sense that they produce elements still in R . The binary operations must satisfy:

$\mathcal{R}_1$: R is an abelian group under the operation $*$.

$\mathcal{R}_2$: $\forall r, s, t \in R$, we have that the left and right distribution laws hold, that is

$$\begin{aligned} r \cdot (s + t) &= (r \cdot s) + (r \cdot t) \\ (s + t) \cdot r &= (s \cdot r) + (t \cdot r). \end{aligned}$$

We call the operation $\cdot$ a *product* and hereafter we will denote the product $\cdot$ by juxtaposition, which means that if we write rs , we mean $r \cdot s$. Hereafter we will denote the binary operation $*$ by $+$ for rings since R is an abelian group under addition. Furthermore, for rings we will denote the additive identity by 0_R .

Definition 4 (Associative ring). Let R be a ring. Then, the ring is an *associative ring* if it also satisfies the following property

$\mathcal{R}_3$: $\forall r, s, t \in R$, it follows that

$$r(st) = (rs)t,$$

which is the *associative* property for the product.

Definition 5 (Zero divisor). In a ring R , an element $r \in R \setminus \{0_R\}$ is a *zero divisor*, provided $\exists s \in R \setminus \{0_R\}$ such that $rs = 0_R$ or $sr = 0_R$.

Definition 6 (Commutative ring). A *commutative ring* is a ring R , which also satisfies the property

$\mathcal{R}_4$: $rs = sr \quad \forall r, s \in R$.

Although groups must have a neutral element, which we denote by 0_G , rings must not have such an element for the second binary operation. We recall the following definition [17].

Definition 7 (Ring with identity). A *ring with identity* is a ring R , which also meets the criteria

$\mathcal{R}_5$: $\exists 1_R \in R$ such that $r1_R = 1_Rr = r \quad \forall r \in R$. We will call 1_R the *multiplicative identity*.

Definition 8 (Unit). In a ring R with identity $1_R \neq 0_R$, an element $u \in R$ is called a *unit* if $\exists r \in R$ such that $ru = ru = 1_R$.

Definition 9 (Field). A *field* is a commutative, associative ring with identity $1_R \neq 0_R$ where each nonzero element is a unit.

If each nonzero element is a unit in a ring, we will refer to the ring as a *division ring* [12].

2.1.1 Vector Spaces and Algebras

Definition 10 (Scalar multiplication). Let V be an abelian group. Furthermore, let F be a field. Define a function $F \times V \rightarrow V$ by $(\alpha, v) \mapsto \alpha v \forall \alpha \in F$ and $\forall v \in V$. We refer to this function as *scalar multiplication*.

Definition 11 (Vector space). Let V be an abelian group, and let F be a field. Then V is called a *vector space* over F if we have an operation of scalar multiplication and provided that for all $\alpha, \beta \in F$ and $u, v \in V$, the following are satisfied:

$$\mathcal{V}_1: \alpha u \in V.$$

$$\mathcal{V}_2: \alpha(\beta u) = (\alpha\beta)u.$$

$$\mathcal{V}_3: (\alpha + \beta)u = \alpha u + \beta u.$$

$$\mathcal{V}_4: \alpha(u + v) = \alpha u + \alpha v.$$

$$\mathcal{V}_5: 1_F u = u.$$

We refer to the elements of F as *scalars* and the elements of V are called *vectors* [12]. An algebra requires, in addition to being a vector space, a *bilinear mapping* [14].

Definition 12 (Linear subspace). Let V be a vector space over a field F . Furthermore, let $W \subseteq V$ and let W be non-empty. Then, W is a *linear subspace* of V if $\forall u, v \in W$ we have $u + v \in W$ and $\alpha u \in W$ for any $\alpha \in F$.

Definition 13 (Span). Let V be a vector space over a field F . Furthermore, let $S = \{v_i | i \in I\}$ be a set of vectors where I is some index set. Then S *span* V if $\forall u \in V$ and for scalars $\alpha_i \in F$ for $i \in I$, we have

$$u = \sum_{i \in I} \alpha_i v_i.$$

We say that a vector u on the form $u = \sum_{i \in I} \alpha_i v_i$ is a linear combination of the vectors v_i .

Definition 14 (Linear independence). Let S be a set of vectors that span the vector space V over a field F . Then, the vectors in S are *linearly independent* if

$$\sum_{i \in I} \alpha_i v_i = 0 \iff \alpha_i = 0 \forall i \in I.$$

Conversely, if there is a different solution such that $\alpha_i \neq 0 \forall i \in I$ to $\sum_{i \in I} \alpha_i v_i = 0$, then the vectors are said to be *linearly dependent*.

Definition 15 (Basis). If $\mathcal{B}$ is a subset of vectors that are linearly independent and span the vector space V over a field F , then $\mathcal{B}$ is a *basis* for V over F .

Any finite-dimensional vector space has a finite basis and any two basiss of a finite-dimensional vector space have the same number of vectors in their respective basis [12].

Definition 16 (Dimensionality). Let V be a finite-dimensional vector space V over a field F . Then, independent of basis, the amount of vectors in $\mathcal{B}$ is the *dimensionality* of V over F .

Definition 17 (Linear mapping). Let V_1 and V_2 be vector spaces over the same field F . Define a function $\phi : V_1 \rightarrow V_2$ by $u \mapsto \phi(u)$ for any $u \in V_1$. This function is an F -linear mapping provided that $\phi(\alpha u + \beta v) = \alpha\phi(u) + \beta\phi(v)$ holds $\forall u, v \in V_1$ and $\forall \alpha, \beta \in F$.

Definition 18 (Bilinear mapping). Let V_1, V_2 and V_3 be vector spaces over the same field F . Then, the mapping $\phi : V_1 \times V_2 \rightarrow V_3$ is a *bilinear mapping* provided that $\forall v_2 \in V_2$ the map $v_1 \mapsto \phi(v_1, v_2)$ is an F -linear map from V_1 to V_3 and also $\forall v_1 \in V_1$ the map $v_2 \mapsto \phi(v_1, v_2)$ is an F -linear map from V_2 to V_3 .

Remark. If $V_3 = F$ in the definition 18, then the bilinear mapping is called a *bilinear form*.

Definition 19 (Trilinear mapping). Let V_1, V_2, V_3 and V_4 be vector spaces over the same field F . Then, the mapping $\phi : V_1 \times V_2 \times V_3 \rightarrow V_4$ is a *trilinear mapping* provided that $\forall v_2 \in V_2$ and $\forall v_3 \in V_3$ the map $v_1 \mapsto \phi(v_1, v_2, v_3)$ is an F -linear map from V_1 to V_4 . Furthermore, $\forall v_1 \in V_1$ and $\forall v_3 \in V_3$ the map $v_2 \mapsto \phi(v_1, v_2, v_3)$ is an F -linear map from V_2 to V_4 and also $\forall v_1 \in V_1$ and $\forall v_2 \in V_2$ the map $v_3 \mapsto \phi(v_1, v_2, v_3)$ is an F -linear map from V_3 to V_4 .

Remark. If $V_4 = F$ in Definition 19, the mapping is called a *trilinear form*.

Definition 20 (Algebra). Let A be a vector space over a field F , and let $A \times A \rightarrow A$ be a bilinear mapping on A , which we refer to as a *product*. Then A is called an *algebra*.

Definition 21 (Unital algebra). An algebra A over a field F is *unital* if $\exists 1_A \in A$ such that $1_A a = a 1_A = a$ for all nonzero $a \in A$.

Note that any algebra is also a ring.

Definition 22 (Subalgebra). Let A be an algebra with the corresponding vector space V over a field F . Then, B is a *subalgebra* of A if W , which is the vector space for B , is a linear subspace of V and if B is closed under the product inherited by A .

Example 2. Consider $\mathbb{R}^3$ as a vector space over $\mathbb{R}$ and let $u = (u_1, u_2, u_3), v = (v_1, v_2, v_3)$ be vectors in $\mathbb{R}^3$ where $u_i, v_i \in \mathbb{R}, i = 1, 2, 3$. Define a mapping $\mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $(u, v) \mapsto u \times v$ by

$$u \times v = (u_2 v_3 - v_2 u_3, \quad v_1 u_3 - u_1 v_3, \quad u_1 v_2 - v_1 u_2).$$

This is a bilinear mapping and thus the vector space with the defined mapping is an algebra over $\mathbb{R}$. This is the ordinary *cross product* [13]. The mappings as a product is an algebra that is neither commutative nor associative.

2.1.2 Mappings, Involutions and *-algebras

Definition 23 (Injective mapping). Let A, B be some sets and define a mapping $f : A \rightarrow B$ by $a \mapsto f(a) \forall a \in A$. If $\forall a_1, a_2 \in A$ we have $f(a_1) = f(a_2) \implies a_1 = a_2$ then the mapping f is *injective*.

Definition 24 (Surjective mapping). Let A, B be some sets and define a mapping $f : A \rightarrow B$ by $a \mapsto f(a) \forall a \in A$. If $f(A) = B$ we call the mapping f *surjective*.

If a mapping is injective and surjective, we call such a mapping a *bijective mapping*.

Definition 25 (Homomorphism). Let A and B be algebras over the field F and define an F -linear mapping $\phi : A \rightarrow B$. Then this mapping is a *homomorphism* if $\phi(a_1 a_2) = \phi(a_1) \phi(a_2)$ holds $\forall a_1, a_2 \in A$.

Definition 26 (Endomorphism). Let $\phi : A \rightarrow B$ be a homomorphism. If $A = B$ we will call such a homomorphism an *endomorphism*.

Definition 27 (Isomorphism). Let $\phi : A \rightarrow B$ be a homomorphism. If ϕ is bijective we call the mapping ϕ an *isomorphism*. If there is an isomorphism between A and B , then A and B are *isomorphic* to each other and write this as $A \cong B$.

Definition 28 (Automorphism). An *automorphism* is a bijective endomorphism.

Definition 29 (Anti-automorphism). Let A be an algebra and $\phi : A \rightarrow A$ be an F -linear bijective mapping. Then ϕ is an *anti-automorphism* if $\phi(a_1 a_2) = \phi(a_2) \phi(a_1)$ holds $\forall a_1, a_2 \in A$.

Definition 30 (Involution). Let A be an algebra over a field F . Define an F -linear mapping $*$: $A \rightarrow A$ by $a \mapsto a^*$. The mapping $*$ is called an *involution* if the following conditions hold:

1. $(a^*)^* = a$.
2. $(ab)^* = b^* a^* \forall a, b \in A$.

Lemma 1. An involution $*$ is an anti-automorphism.

Proof. Let everything be as in Definition 30. We must show that $*$ is a bijective mapping. For injectivity, we have

$$\text{If } a^* = b^* \text{ then } (a^*)^* = (b^*)^* \iff a = b.$$

Hence $*$ is injective. To show surjectivity, we have $\forall a \in A$ that $a = b^*$ for some $b \in A$. This is $b = a^*$. Therefore $*$ is both injective and surjective, making the mapping bijective. Therefore an involution is an anti-automorphism. This completes the proof. $\square$

Note that an involution is an anti-automorphism.

Definition 31 (*-algebra). An algebra A over a field F equipped with an involution $*$ is called a **-algebra* (star algebra).

Definition 32 (*-algebra isomorphism). Let $\phi : A \rightarrow B$ be an isomorphism such that $\phi(a^*) = \phi(a)^*$. Then ϕ is called a **-algebra isomorphism*, and A and B are said to be *isomorphic as *-algebras*, written $A \cong^* B$.

2.1.3 Normed Vector Spaces and Algebras

Definition 33 (Inner product). Let V be a vector space over the real field $\mathbb{R}$ equipped with a bilinear form $V \times V \rightarrow \mathbb{R}$ by $(u, v) \mapsto \langle u|v \rangle$ where $u, v \in V$ and $\alpha \in \mathbb{R}$. Then the bilinear form is called an *inner product* if $\forall u, v, w \in V$ and $\alpha \in \mathbb{R}$ the following are satisfied:

1. $\langle u|v \rangle = \langle v|u \rangle$.
2. $\langle u|u \rangle \geq 0$ and $\langle u|u \rangle = 0 \iff u = 0$.

If V is a vector space with an inner product, the vector space is called an *inner product space*.

Definition 34 (Absolute value). Define a function $|\cdot| : \mathbb{C} \rightarrow \mathbb{R}$ by $(a, b) \mapsto \sqrt{a^2 + b^2}$. This is the *absolute value* of a complex number.

Remark. The above definition also holds for $|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$ by letting $b = 0$.

Definition 35 (Norm). Let V be a vector space over the field $\mathbb{R}$ or $\mathbb{C}$, which we will refer to as F , and define a function $\|\cdot\| : V \rightarrow \mathbb{R}$. If $\forall u, v \in V$ and $\forall \alpha, \beta \in F$ we have

1. $\|u\| \geq 0$ and $\|u\| = 0$ iff $u = 0$.
2. $\|\alpha u\| = |\alpha| \|u\|$.
3. $\|u + v\| \leq \|u\| + \|v\|$,

then the function is referred to as a *norm*.

If V is a vector space over a field F equipped with a norm, then V is called a *normed vector space* [15]. Furthermore, each inner product induces a norm in a natural way.

Definition 36 (Induced norm). Let V be an inner product space over a field F . Then the *induced norm* is

$$\|u\|^2 = \langle u|u \rangle \quad \forall u \in V.$$

Theorem 1 (Cauchy–Schwarz inequality). *Let V be an inner product space over the field $\mathbb{R}$ or $\mathbb{C}$. Then, any vectors $u, v \in V$ satisfies*

$$|\langle u|v \rangle| \leq \|u\| \|v\|.$$

Proof. Proof is omitted. See e.g. [13]. □

We may now extend the idea of a normed vector space to a normed algebra.

Definition 37 (Normed algebra). Let A be an algebra that is also a normed vector space over a field F . If the norm also satisfies

$$\|ab\| \leq \|a\| \|b\| \quad \forall a, b \in A,$$

then A is called a *normed algebra*.

Definition 38 (Absolute valued algebra). Let A be a normed algebra over a field F . If the norm also satisfies

$$\|ab\| = \|a\| \|b\| \quad \forall a, b \in A,$$

then A is called an *absolute valued algebra*.

Similarly to the definition of a division ring, we may define a division algebra in a similar way [21].

Definition 39 (Division algebra). A *division algebra* is an algebra A over a field F that contains no zero divisors and where every nonzero element is an unit.

We end this section with a link between absolute valued algebras and division algebras. As described by Albert [3], A must be a division algebra if it is an absolute valued algebra. We will call such an algebra a *absolute valued division algebra*.

Proposition 1 (A is a division algebra if it is an absolute valued algebra). *If A is an absolute valued algebra over a field F , then it is a division algebra.*

Proof. We must only show that A is a division algebra. Let $a, b \in A$. Then we have

$$\text{If } a \neq 0 \text{ and } b \neq 0 \text{ then } \|a\| > 0 \text{ and } \|b\| > 0 \text{ thus } \|ab\| = \|a\| \|b\| > 0 \implies ab \neq 0.$$

Thereby A contains no zero divisors since the norm is only zero if $a = 0$ or $b = 0$. Therefore, each nonzero element is an unit and the algebra A is an absolute valued division algebra. This completes the proof. $\square$

2.1.4 Direct Sums and Products

Definition 40 (Cartesian product of sets). Let $I = \{1, 2, \dots, n\}$. Let S_i be sets where $i \in I$. Then the set of all n -tuples $(s_1, s_2, \dots, s_n)$ where $s_i \in S_i \forall i \in I$ is the Cartesian product of sets. The Cartesian product is denoted by

$$S_1 \times S_2 \times \dots \times S_n \text{ or } \prod_{i=1}^n S_i.$$

Proposition 2 (Direct sum of vector spaces). *Let $I = \{1, 2, \dots, n\}$. Let V_i with $\forall i \in I$ be vector spaces over the same field F . Then the Cartesian product of sets $\prod_{i=1}^n V_i$ is a vector space and is called the direct product of vector spaces, denoted by $\bigoplus_{i=1}^n V_i$, if we define addition and scalar multiplication as:*

$$1. \text{ Addition: } (u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n),$$

$$2. \text{ Scalar multiplication: } \alpha(u_1, u_2, \dots, u_n) = (\alpha u_1, \alpha u_2, \dots, \alpha u_n)$$

$$\forall (u_1, u_2, \dots, u_n) \in \bigoplus_{i=1}^n V_i \text{ and } \forall \alpha \in F.$$

Proof. We need to prove that all axioms in Definition 11 hold. First let $\alpha, \beta \in F$ and $(u_1, u_2, \dots, u_n), (v_1, v_2, \dots, v_n) \in \bigoplus_{i=1}^n V_i$.

$\mathcal{V}_1$: We ought to prove $\alpha(u_1, u_2, \dots, u_n) \in \bigoplus_{i=1}^n V_i$.

$\alpha(u_1, u_2, \dots, u_n) = (\alpha u_1, \alpha u_2, \dots, \alpha u_n) \in \bigoplus_{i=1}^n V_i$ as $\alpha u_i \in V_i$ for each component and these are already vector spaces. This completes the first part of the whole proof.

$\mathcal{V}_2$: We have to prove $\alpha(\beta(u_1, u_2, \dots, u_n)) = (\alpha\beta)(u_1, u_2, \dots, u_n)$.

$\alpha(\beta(u_1, u_2, \dots, u_n)) = \alpha(\beta u_1, \beta u_2, \dots, \beta u_n) = (\alpha\beta u_1, \alpha\beta u_2, \dots, \alpha\beta u_n) = (\alpha\beta)(u_1, u_2, \dots, u_n)$. This completes the proof for this axiom.

$\mathcal{V}_3$: We must prove $(\alpha + \beta)(u_1, u_2, \dots, u_n) = \alpha(u_1, u_2, \dots, u_n) + \beta(u_1, u_2, \dots, u_n)$.

$$\begin{aligned} (\alpha + \beta)(u_1, u_2, \dots, u_n) &= ((\alpha + \beta)u_1, (\alpha + \beta)u_2, \dots, (\alpha + \beta)u_n) = \\ &= (\alpha u_1 + \beta u_1, \alpha u_2 + \beta u_2, \dots, \alpha u_n + \beta u_n) = (\alpha u_1, \alpha u_2, \dots, \alpha u_n) + (\beta u_1, \beta u_2, \dots, \beta u_n) = \\ &= \alpha(u_1, u_2, \dots, u_n) + \beta(u_1, u_2, \dots, u_n). \end{aligned}$$

This concludes this part of the proof.

$\mathcal{V}_4$: Here we have to prove $\alpha((u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n)) = \alpha(u_1, u_2, \dots, u_n) + \alpha(v_1, v_2, \dots, v_n)$.

$$\begin{aligned} \alpha((u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n)) &= \alpha(u_1 + v_1, u_2 + v_2, \dots, u_n + v_n) = \\ &= (\alpha(u_1 + v_1), \alpha(u_2 + v_2), \dots, \alpha(u_n + v_n)) = (\alpha u_1 + \alpha v_1, \alpha u_2 + \alpha v_2, \dots, \alpha u_n + \alpha v_n) = \\ &= (\alpha u_1, \alpha u_2, \dots, \alpha u_n) + (\alpha v_1, \alpha v_2, \dots, \alpha v_n) = \alpha(u_1, u_2, \dots, u_n) + \alpha(v_1, v_2, \dots, v_n). \end{aligned}$$

Therefore we have shown this part of the proof.

$\mathcal{V}_5$: We must prove $1_F(u_1, u_2, \dots, u_n) = (u_1, u_2, \dots, u_n)$.

$1_F(u_1, u_2, \dots, u_n) = (1_F u_1, 1_F u_2, \dots, 1_F u_n) = (u_1, u_2, \dots, u_n)$, which completes the proof of this axiom.

All axioms hold and thus, this is a vector space. This completes the proof. $\square$

2.2 The Commuter, Nuclei and Center of an Algebra

Definition 41 (Commutator). Let A be an algebra over a field F . Define a bilinear mapping $A \times A \rightarrow A$ by $[a, b] = ab - ba$. This is the *commutator map*.

Definition 42 (Commuter). Let A be an algebra over a field F . Define the *commuter* to be the set $C(A) = \{a \in A : [a, b] = 0 \quad \forall b \in A\}$.

Definition 43 (Associator). Let A be an algebra over a field F . Define a trilinear mapping $A \times A \times A \rightarrow A$ by $[a, b, c] = (ab)c - a(bc)$. This is the *associator map*.

Lemma 2 (Associator identity). *Let A be an algebra over a field F . Furthermore, let $a, b, c, d \in A$. Then the associator satisfies the associator identity,*

$$a[b, c, d] + [a, b, c]d = [ab, c, d] - [a, bc, d] + [a, b, cd].$$

Proof.

$$\begin{aligned} a[b, c, d] + [a, b, c]d &= a((bc)d - b(cd)) + ((ab)c - a(bc))d \\ &= a((bc)d) - a(b(cd)) + ((ab)c)d - (a(bc))d \\ [ab, c, d] - [a, bc, d] + [a, b, cd] &= ((ab)c)d - (ab)(cd) - (a(bc))d + a((bc)d) + \\ &\quad (ab)(cd) - a(b(cd)) = ((ab)c)d - (a(bc))d + a((bc)d) - a(b(cd)) \end{aligned}$$

By observing the above expression, we see that both sides are equal and hence the identity holds. This completes the proof. $\square$

Definition 44 (Alternative). Let A be an algebra over a field F . The algebra is said to be alternative if:

1. $a(ab) = (aa)b$,
2. $(ba)a = b(aa)$ and
3. $(ab)a = a(ba)$ holds $\forall a, b \in A$

Artin [5] proved that we must only check two of the three equations in Definition 44 in order to prove that an algebra is alternative as the third is then implied.

We may define the left, middle and right *nuclei* to be all the elements in an algebra where the associator vanishes by holding any of the three arguments fixed and letting the domain be all possible elements in the algebra.

Definition 45 (Left nuclei). Let A be an algebra over a field F . Define the *left nuclei* to be the set $N_l(A) = \{a \in A : [a, b, c] = 0 \quad \forall b, c \in A\}$.

We may define the middle nuclei and the right nuclei in a similar way.

Definition 46 (Middle nuclei). Let A be an algebra over a field F . Define the *middle nuclei* to be the set $N_m(A) = \{b \in A : [a, b, c] = 0 \quad \forall a, c \in A\}$.

Definition 47 (Right nuclei). Let A be an algebra over a field F . Define the *right nuclei* to be the set $N_r(A) = \{c \in A : [a, b, c] = 0 \quad \forall a, b \in A\}$.

Definition 48 (Nucleus). Let $N_l(A)$, $N_m(A)$ and $N_r(A)$. We define the *nucleus* to be the set $N(A) = N_l(A) \cap N_m(A) \cap N_r(A)$.

Proposition 3. $N_l(A)$, $N_m(A)$, $N_r(A)$ and $N(A)$ of an algebra A over a field F are associative subalgebras of A .

Proof. We will prove that $N_l(A)$ is an associative subalgebra of A . First, we must show that $N_l(A)$ is a linear subspace of A . Let $a_1, a_2 \in N_l(A)$ and $b, c \in A$. Then $a_1 + a_2$ must be in $N_l(A)$.

$$\begin{aligned} [a_1 + a_2, b, c] &= ((a_1 + a_2)b)c - (a_1 + a_2)(bc) = (a_1b)c + (a_2b)c - a_1(bc) - a_2(bc) \\ &= (a_1b)c - a_1(bc) + (a_2b)c - a_2(bc) = [a_1, b, c] + [a_2, b, c] = 0 + 0 = 0. \end{aligned}$$

Therefore $a_1 + a_2 \in N_l(A)$. Furthermore, let $\alpha \in F$. We must show $\alpha a_1 \in N_l(A)$.

$$\begin{aligned} [\alpha a_1, b, c] &= (\alpha a_1 b)c - \alpha a_1(bc) = \alpha(a_1b)c - \alpha a_1(bc) \\ &= \alpha((a_1b)c - a_1(bc)) = \alpha[a_1, b, c] = \alpha 0 = 0. \end{aligned}$$

Therefore, $\alpha a_1 \in N_l(A)$ and $N_l(A)$ is a linear subspace of the vector space of A . To show that $N_l(A)$ is a subalgebra of A we must also show $a_1 a_2 \in N_l(A)$. We will do so by using the associator identity.

$$\begin{aligned} a_1[a_2, b, c] + [a_1, a_2, b]c &= [a_1 a_2, b, c] - [a_1, a_2 b, c] + [a_1, a_2, bc] \iff \\ a_1 0 + 0c &= [a_1 a_2, b, c] - 0 + 0 \iff [a_1 a_2, b, c] = 0 \implies a_1 a_2 \in N_l(A). \end{aligned}$$

Since $N_l(A)$ is a linear subspace of A and closed under multiplication, $N_l(A)$ is a subalgebra of A . Furthermore, $N_l(A)$ is associative since if $a_1, a_2, a_3 \in A$ then $[a_1, a_2, a_3] = 0$ since $a_1 \in N_l(A)$. The proofs for $N_m(A)$ and $N_r(A)$ are similar and omitted. Since $N(A)$ is the intersection of the left, middle and right nuclei, $N(A)$ is associative as well. This completes the proof. $\square$

Definition 49 (Center). Let A be an algebra over a field F . Then, the *center* of A is the set $Z(A) = C(A) \cap N(A)$.

$N(A)$ is an associative subalgebra of A and therefore $Z(A)$ is a commutative and associative subalgebra of A .

2.3 Relations and Orderings

Definition 50 (Binary relation). Let S_1 and S_2 be sets and construct the Cartesian product $S_1 \times S_2$. A *binary relation* from S_1 to S_2 , denoted by R , is a subset of $S_1 \times S_2$ of all the ordered pairs (s_1, s_2) where $s_1 \in S_1$ and $s_2 \in S_2$ such that $s_1 R s_2$. We say that s_1 is related to s_2 through R .

Any binary relation gives rise to a subset of $S_1 \times S_2$ given by $R^* = \{(s_1, s_2) : s_1 R s_2\}$. In contrast, any subset $R^* \subseteq S_1 \times S_2$ gives rise to a relation [15]. A relation from S_1 to itself is a relation in S_1 .

Definition 51 (Cardinality). Let S be a set. The *cardinality* of S is the number of elements in S . We denote the cardinality by $\#S$.

Definition 52 (Irreflexive property). Let R be a relation in S . If $\forall s \in S$ we have that $s \not R s$, then the relation is called *irreflexive*.

Remark. If $s R s$ above, then the relation is said to be *reflexive*.

Definition 53 (Transitive property). Let R be a relation in S . If $s_1 R s_2$ and $s_2 R s_3$, then $s_1 R s_3$ $\forall s_1, s_2, s_3 \in S$, then the relation is called *transitive*.

Definition 54 (Asymmetric property). Let R be a relation in S . If $s_1 R s_2$ and $s_2 \not R s_1$ $\forall s_1, s_2 \in S$, then we call the relation *asymmetric*.

Definition 55 (Anti-symmetric property). Let R be a relation in S . Furthermore, if $s_1 R s_2$ and $s_2 R s_1$ then we have $s_1 = s_2$ $\forall s_1, s_2 \in S$. Such a relation is called *anti-symmetric*.

Definition 56 (Partial order). Let $\preceq$ be a relation in S . Then $\preceq$ is called a *partial order* if $\prec$ is reflexive, transitive and anti-symmetric.

Definition 57 (Strict partial order). Let $\prec$ be a relation in S . Then $\prec$ is called a *strict partial order* if $\prec$ is irreflexive, transitive and asymmetric.

We will refer to (strictly) *partially ordered set* as a set with a (strict) partial order and denote this by $(\langle S, \prec \rangle)$ $(\langle S, \preceq \rangle)$.

Definition 58 (Total order). Let $\langle S, \preceq \rangle$. If $\forall s_1, s_2 \in S$ we either have $s_1 \preceq s_2$ or $s_2 \preceq s_1$, then S is called a *totally ordered set*.

Remark. The above definition also holds for a strictly partially ordered set S .

We will denote a totally ordered set by $\langle S, \prec \rangle^T$ or $\langle S, \preceq \rangle^T$ depending on the partial order. For a strictly partially ordered set we will utilize the common-used symbol $<$ for our relation and for a partially ordered set we will utilize the symbol $\leq$. We may connect the idea of an ordered set with an ordered field [18].

Definition 59 (Positive elements). Let F be a field and let $E \subseteq F$ and let E be non-empty. We call E the set of *positive elements* if for any $\alpha \in E$ we have $0 \leq \alpha$.

Definition 60 (Ordered field). Let F be a field and let E be the positive elements of F . Then F is an *ordered field* if it satisfies

1. $\forall \alpha, \beta \in E$ we have $\alpha + \beta \in E$ and $\alpha\beta \in E$.
2. If $\alpha \in F \setminus \{0\}$ then either $\alpha \in E$ or $-\alpha \in E$ but never both.

Remark. By definition, if F is an ordered field under any relation then the set F is a totally ordered set.

Example 3. Let F be an ordered field with positive elements E . Furthermore, let $\langle F, < \rangle^T$. Then, for any $\alpha, \beta \in \langle F, < \rangle^T$ we have $\alpha < \beta$ or $\beta < \alpha$. Without loss of generality, assume $\alpha < \beta$, then we have $0 < \beta - \alpha$ which means $\beta - \alpha \in E$. Also, without loss of generality, assume $0 < \alpha < \beta$, then $0 < \alpha^2 < \alpha\beta$ so $\alpha\beta \in E$. The case where we allow α to be negative can be treated in a similar fashion, as can the whole example with the binary relation $\leq$.

By Example 3 we can observe that every ordered field is also an ordered set [18]. Furthermore, it is evident that $\alpha < \beta \implies \beta - \alpha$ and $\alpha \leq \beta \implies \alpha = \beta$ or $\alpha < \beta$.

Theorem 2 ($\mathbb{R}$ is an ordered field). *$\mathbb{R}$ is an ordered field.*

Proof. By assumption, if $\mathbb{R}$ is an order field, $\mathbb{R}$ must also be an ordered set. Let $\alpha, \beta, \gamma \in \mathbb{R}$. We have $\alpha < \beta$, $\beta < \alpha$ or $\alpha = \beta$. Furthermore, if $\alpha < \beta$ and $\beta < \gamma$ then $\alpha < \gamma$. Transitivity also holds by replacing $<$ with $\leq$. Now, we have that if $\alpha < \beta$ then $0 < \beta - \alpha \in E$. Likewise if $\alpha = \beta$ we have $0 \leq 0 \in E$. Thus, for $<$ we have that $E = \mathbb{R}^+$ and for $\leq$ we have that $E = \mathbb{R}^+ \cup \{0\}$. In either case, for any $\alpha, \beta \in E$ $\alpha + \beta \in E$ and $\alpha\beta \in E$. Regardless of relation, let $\alpha \neq 0$. If $\alpha \in E$, then $-\alpha \notin E$ as E does not contain negative numbers. Hence $\mathbb{R}$ is an ordered field under both $<$ and $\leq$. This completes the proof. $\square$

Chapter 3

The Cayley–Dickson Construction

3.1 The Standard Involution and Construction

The standard Cayley–Dickson construction is a bit different from what will be given here, as we will use the more general version introduced by Albert [3][4].

Definition 61 (Cayley–Dickson construction). Let A be a $*$ -algebra over a field F and let $\mu \in F \setminus \{0\}$. Furthermore, let $A \oplus A$ be the direct sum of vector spaces. Define a product of elements in $A \oplus A$ by $(a, b)(c, d) = (ac + \mu d^*b, da + bc^*)$ where $a, b, c, d \in A$. This process, which is the *Cayley–Dickson construction*, creates a new algebra. We will denote the new algebra generated by the construction by $\text{Cay}(A, \mu)$.

Remark. If $\mu = -1$, then this is the standard product discovered by Dickson and Cayley.

The following involution is the classical involution by Dickson and Cayley.

Theorem 3 (The standard involution for Cayley–Dickson). *Let A be a $*$ -algebra and define a mapping $*$: $A \oplus A \rightarrow A \oplus A$ by $(a, b)^* = (a^*, -b)$. Then $*$ is an involution on $\text{Cay}(A, \mu)$.*

Proof. We first prove that the first part in Definition 30 holds. Let $(a, b) \in A \oplus A$. Then, we have

$$((a, b)^*)^* = (a^*, -b)^* = ((a^*)^*, -(-b)) = (a, b).$$

Hence the first requirement holds. Now for the second part, we let $(a, b), (c, d) \in A \oplus A$.

$$\begin{aligned} ((a, b)(c, d))^* &= (ac + \mu d^*b, da + bc^*)^* = ((ac + \mu d^*b)^*, -da - bc^*) \\ &= ((ac)^* + (\mu d^*b)^*, -da - bc^*) = (c^*a^* + b^*\mu d^*, -da - bc^*) = (c^*a^* + \mu b^*d, -da - bc^*). \\ (c, d)^*(a, b)^* &= (c^*, -d)(a^*, -b) = (c^*a^* + \mu(-b)^*(-d), -bc^* - d(a^*)^*) \\ &= (c^*a^* + \mu b^*d, -da - bc^*). \end{aligned}$$

Therefore the second part also holds. We must also show that this mapping is F -linear. First,

let $\alpha, \beta \in F$. Then, we have

$$\begin{aligned} (\alpha(a, b) + \beta(c, d))^* &= (\alpha a + \beta c, \alpha b + \beta d)^* = ((\alpha a + \beta c)^*, -(\alpha b + \beta d)) = \\ &= (\alpha a^* + \beta c^*, -(\alpha b + \beta d)) = (\alpha a^*, -\alpha b) + (\beta c^*, -\beta d) = \alpha(a^*, -b) + \beta(c^*, -d) \\ &= \alpha(a, b)^* + \beta(c, d)^*. \end{aligned}$$

Hence, the mapping is F -linear and we have proven this is an involution. $\square$

Using $\text{Cay}(A, \mu)$ on $A = \mathbb{R}$ with $\mu = -1$ and the involution given in Theorem 3, we will generate the complex numbers ($\mathbb{C}$), the quaternions ($\mathbb{H}$) and the octonions ($\mathbb{O}$) in the first 3 doublings. If we instead use $A = \mathbb{R}$ with $\mu = 1$, then we obtain the so-called split versions. These are called the split complex numbers ($\mathbb{C}'$), the split quaternions ($\mathbb{H}'$), and the split octonions ($\mathbb{O}'$). As with the standard choice of $\mu = -1$, the split numbers have all been well studied.

Proposition 4 (One involution on $\mathbb{R}$). *The only involution on the algebra $\mathbb{R}$ over $\mathbb{R}$ is the identity mapping $*$: $\mathbb{R} \rightarrow \mathbb{R}$ by $a \mapsto a^* = a$.*

Proof. This ought to be determined by the mapping of 1. First, we have

$$1^* = (1 \cdot 1)^* = (1)^*(1)^* = ((1)^*)^2 \iff (1)^* = 1 \text{ or } (1)^* = 0.$$

If $(1)^* = 0$, then we have problems as $(a \cdot 1)^* = (1)^*(a)^* = 0(a)^* = 0 \forall a \in \mathbb{R}$ so this is not a one-to-one mapping. Therefore, we consider $(1)^* = 1$, instead. Since $*$ is $\mathbb{R}$ -linear, we have

$$(a)^* = a(1)^* = a \cdot 1 = a.$$

Therefore $*$ is the identity mapping. The identity mapping is trivially an involution. This completes the proof. $\square$

3.1.1 The Complex Numbers $\mathbb{C}$

We will examine the first three algebras obtained through the Cayley–Dickson construction by starting with $\mathbb{R}$, the identity involution on $\mathbb{R}$ and utilizing the standard choice of $\mu = -1$. We start with $\text{Cay}(\mathbb{R}, -1)$ with the involution on $\mathbb{R} \oplus \mathbb{R}$ given in Theorem 3. We denote the algebra with the particular choice of involution by $\text{Cay}_-(\mathbb{R}, -1)$. By this process, we should obtain the complex numbers with ordinary complex conjugation. We have $(a, b) \in \mathbb{R} \oplus \mathbb{R}$. We may prove that this algebra is commutative and associative.

Proposition 5. *$\text{Cay}(\mathbb{R}, -1)$ is a commutative algebra.*

Proof. Let $(a, b), (c, d) \in \text{Cay}(\mathbb{R}, -1)$. Then, by the commutator, we have

$$\begin{aligned} [(a, b), (c, d)] &= (a, b)(c, d) - (c, d)(a, b) = (ac - d^*b, da + bc^*) - (ca - b^*d, bc + dca^*) \\ &= (ac - db, da + bc) - (ca - bd, bc + da) = (ac - db - ca + bd, da + bc - da - bc) = (0, 0) \\ &\implies (a, b)(c, d) = (c, d)(a, b). \end{aligned}$$

The commutator is zero and hence the proof follows. We have utilized that a, b, c, d are real numbers and $a^* = a$ for all real numbers. This completes the proof. $\square$

Proposition 6. $\text{Cay}(\mathbb{R}, -1)$ is an associative algebra.

Proof. Let $(a, b), (c, d), (e, f) \in \text{Cay}(\mathbb{R}, -1)$. Then, by using the associator, we have

$$\begin{aligned}
[(a, b), (c, d), (e, f)] &= \left((a, b)(c, d) \right)(e, f) - (a, b) \left((c, d)(e, f) \right) = \\
&= (ac - d^*b, da + bc^*)(e, f) - (a, b)(ce - f^*d, fc + de^*) = (ac - db, da + bc)(e, f) \\
&\quad - (a, b)(ce - fd, fc + de) = ((ac - db)e - f^*(da + bc), f(ac - db) + (da + bc)e^*) \\
&\quad - (a(ce - fd) - (fc + de)^*b, (fc + de)a + b(ce - fd)^*) \\
&= (ace - dbe - fda - fbc, fac - fdb + dae + bce) \\
&\quad - (ace - afd - fcb - deb, fca + dea + bce - bfd) = (0, 0) \\
&\implies \left((a, b)(c, d) \right)(e, f) = (a, b) \left((c, d)(e, f) \right).
\end{aligned}$$

The associator is zero and hence the proof follows. We have utilized that $a, b, \dots, f \in \mathbb{R}$ and multiplication in $\mathbb{R}$ is commutative. This concludes the proof. $\square$

Recall that the complex numbers may be written in the form $z_1 = a + bi$ where i is the imaginary unit. If $z_2 = c + di$ is yet another complex number, we may multiply these two complex numbers as

$$z_1 z_2 = (a + bi)(c + di) = (ac - db) + (da + bc)i. \quad (3.1)$$

We take a look at the complex numbers through the Cayley–Dickson construction now. By previous statements, we should have $\mathbb{C} = \text{Cay}(\mathbb{R}, -1)$. Let $(a, b), (c, d) \in \text{Cay}(\mathbb{R}, -1)$ and multiply them

$$(a, b)(c, d) = (ac - d^*b, da + bc^*) = (ac - db, da + bc). \quad (3.2)$$

Since $a, b, c, d \in \mathbb{R}$, we have $d^* = d$ and $c^* = c$. By comparing equation (3.1) with equation (3.2), we observe that these equations describe the same thing as the pair (a, b) can be thought of as $a + bi$. This is the factual definition. We can prove this fact now.

Proposition 7. $\text{Cay}_-(\mathbb{R}, -1) \stackrel{*}{\cong} \mathbb{C}$.

Proof. Let $\phi : \text{Cay}(\mathbb{R}, -1) \rightarrow \mathbb{C}$ be a mapping defined by $(a, b) \mapsto a + bi$. We ought to prove that ϕ is a $*$ -algebra isomorphism. First, we show that ϕ is $\mathbb{R}$ -linear. Let $(a, b), (c, d) \in \text{Cay}(\mathbb{R}, -1)$ and $\alpha, \beta \in \mathbb{R}$. Then, $\phi(\alpha(a, b) + \beta(c, d)) = \phi((\alpha a + \beta c, \alpha b + \beta d)) = (\alpha a + \beta c) + (\alpha b + \beta d)i = \alpha(a + bi) + \beta(c + di) = \alpha\phi((a, b)) + \beta\phi((c, d))$. Hence, ϕ is a $\mathbb{R}$ -linear mapping. Furthermore, we have $\phi((a, b)(c, d)) = \phi((ac - db, da + bc)) = (ac - db) + (da + bc)i$. Additionally, we have $\phi((a, b))\phi((c, d)) = (a + bi)(c + di) = (ac - db) + (da + bc)i$. Thus, $\phi((a, b), (c, d)) = \phi((a, b))\phi((c, d))$. Hereby, ϕ is a homomorphism. We can show that this mapping is injective. $\phi((a, b)) = \phi((c, d)) \iff a + bi = c + di$, hence $a = c$ and $b = d$, which means $(a, b) = (c, d)$. Surjectivity is immediate. Lastly, we have $\phi((a, b)^*) = \phi((a, -b)) = a - bi = (a + bi)^* = \phi((a, b))^*$. Therefore, it follows that ϕ is a $*$ -algebra isomorphism and hence $\text{Cay}(\mathbb{R}, -1) \stackrel{*}{\cong} \mathbb{C}$. $\square$

Proposition 8 (Representation of i). $(0, 1) \in \text{Cay}(\mathbb{R}, -1)$ is represented by $i \in \mathbb{C}$ and $(1, 0) \in \text{Cay}(\mathbb{R}, -1)$ is represented by $1 \in \mathbb{C}$. Therefore, $(0, 1)(0, 1) = -(1, 0) = (-1, 0)$.

Proof. The proof is straightforward. Take the mapping ϕ in proposition 7, then $\phi((1, 0)) = 1 + 0i = 1$ and also $\phi((0, 1)) = 0 + 1i = i$. Therefore, it follows that $\phi((0, 1)(0, 1)) = \phi((0, 1))\phi((0, 1)) = i^2 = -1 = (-1, 0)$. This completes the proof. $\square$

One may show that $\mathbb{C}$ is a field although unlike $\mathbb{R}$ it is not an ordered field [18].

Lemma 3 ($\mathbb{C}$ is not an ordered field under the relation $<$).

Proof. We will prove this fact by contradiction. If $\mathbb{C}$ were to be an ordered field, then $\langle \mathbb{C}, < \rangle^T$. If $0 < i$, then we would have $0 < i \iff i0 < i^2 \iff 0 < -1$. But this cannot be the case. If instead we have $i < 0$, we would obtain $0 < -i \iff -i0 < i^2 \iff 0 < -1$. Thus, $i < 0$ and $0 < i$ are not possible, since the partial order must satisfy either this is a contradiction. Hence $\mathbb{C}$ cannot be an ordered field under $<$. This completes the proof. $\square$

Theorem 4 ($\mathbb{C}$ is not an ordered field).

Proof. We already know that $\mathbb{C}$ is not an ordered field under a strict partial order. Hence, we must examine the relation $\leq$. We may have $0 \leq i$. By Lemma 3 this would mean $0 = i \iff 0i = i^2 \iff 0 = -1$. This is a contradiction. Hence $\mathbb{C}$ is not an ordered field, this completes the proof. $\square$

Proposition 9 ($\text{Cay}(\mathbb{R}, -1)$ is an inner product space.). Define a bilinear form $\langle \cdot | \cdot \rangle : \text{Cay}(\mathbb{R}, -1) \rightarrow \mathbb{R}$ by $\langle a | b \rangle = \frac{1}{2}(a^*b + b^*a)$. Then, this mapping is an inner product and $\text{Cay}(\mathbb{R}, -1)$ is an inner product space.

Proof. Let $a := (a_1, a_2), b := (b_1, b_2)$ with $a_1, a_2, b_1, b_2 \in \mathbb{R}$. Then, we have

$$\begin{aligned} \langle a | b \rangle &= \frac{1}{2} \left((a_1, -a_2)(b_1, b_2) + (b_1, -b_2)(a_1, a_2) \right) \\ &= \frac{1}{2} \left((a_1b_1 + b_2a_2, b_2a_1 - a_2b_1) + (b_1a_1 + a_2b_2, a_2b_1 - b_2a_1) \right) \\ &= \frac{1}{2} (2a_1b_1 + 2a_2b_2, 0) = (a_1b_1 + a_2b_2, 0). \end{aligned}$$

Evidently, the equation above is only 0 is $a_1 = a_2 = b_1 = b_2 = 0$. Symmetry is immediate as the components are real and commutative. The inner product is non-negative if $a = b$. We also need to prove that this mapping is bilinear.

$$\begin{aligned} \langle \alpha_1 a + \alpha_2 b | \beta_1 c + \beta_2 d \rangle &= \frac{1}{2} \left((\alpha_1 a + \alpha_2 b)^* (\beta_1 c + \beta_2 d) + (\beta_1 c + \beta_2 d)^* (\alpha_1 a + \alpha_2 b) \right) \\ &= \frac{1}{2} \left((\alpha_1 a^* + \alpha_2 b^*) (\beta_1 c + \beta_2 d) + (\beta_1 c^* + \beta_2 d^*) (\alpha_1 a + \alpha_2 b) \right) \\ &= \frac{1}{2} \left(\alpha_1 a^* \beta_1 c + \alpha_1 a^* \beta_2 d + \alpha_2 b^* \beta_1 c + \alpha_2 b^* \beta_2 d + \beta_1 c^* \alpha_1 a + \beta_1 c^* \alpha_2 b + \beta_2 d^* \alpha_1 a + \beta_2 d^* \alpha_2 b \right) \\ &= \frac{1}{2} \left(\alpha_1 \beta_1 (a^* c + c^* a) + \alpha_1 \beta_2 (a^* d + d^* a) + \alpha_2 \beta_1 (b^* c + c^* b) + \alpha_2 \beta_2 (b^* d + d^* b) \right) \\ &= \alpha_1 \beta_1 \langle a | c \rangle + \alpha_1 \beta_2 \langle a | d \rangle + \alpha_2 \beta_1 \langle b | c \rangle + \alpha_2 \beta_2 \langle b | d \rangle. \end{aligned}$$

This mapping satisfies all criteria for being an inner product and $\text{Cay}(\mathbb{R}, -1)$ is therefore an innerproduct space. $\square$

The standard norm on $\mathbb{C}$ is $\|z\| = \sqrt{zz^*} = \sqrt{(a+bi)(a-bi)} = \sqrt{a^2+b^2}$ (see e.g. [20]). Since $\phi : \text{Cay}_-(\mathbb{R}, -1) \rightarrow \mathbb{C}$ is a $*$ -algebra isomorphism, we have that $\phi^{-1} : \mathbb{C} \rightarrow \text{Cay}_-(\mathbb{R}, -1)$ is a $*$ -algebra isomorphism. We can therefore show that we get exactly the same norm with our ordered pair through the Cayley–Dickson construction.

Proposition 10 (A norm for $\text{Cay}(\mathbb{R}, -1)$). *Since $\|\cdot\|$ is a norm on $\mathbb{C}$, we have that $\phi^{-1}(\|\cdot\|)$ is a norm on $\text{Cay}(\mathbb{R}, -1)$.*

Proof. First let $z = a+bi \in \mathbb{C}$. Then, we have $\phi^{-1}(\|z\|) = \phi^{-1}(\sqrt{a^2+b^2}) = (\sqrt{a^2+b^2}, 0)$. $\square$

Note that the norm is given by the square root of the inner product in Proposition 9 and the norm is therefore induced by the inner product.

Proposition 11. *$\text{Cay}(\mathbb{R}, -1)$ is a division algebra.*

Proof. We must show that each element is an unit and that there are no zero divisors. First, we prove which form the unit must be on. Let $(a, b) \in \text{Cay}(\mathbb{R}, -1)$. Then, (c, d) is a unit if $(a, b)(c, d) = (c, d)(a, b) = (1, 0)$. We have

$$(a, b)(c, d) = (ac - db, da + bc) \implies \begin{cases} ac - db = 1 \\ da + bc = 0. \end{cases} \quad (3.3)$$

Since, $a, b, c, d \in \mathbb{R}$, we may divide and solve for d in the second equation to obtain $d = \frac{-bc}{a}$ where $a \neq 0$. We put this into the first equation and obtain $c = \frac{a}{a^2+b^2}$. The inverse for (a, b) is therefore $(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2})$ if $a \neq 0$. If $a = 0$, then the fractions are defined and the units are the same. We must now show that there are no zero divisors, we do this by contradiction. We must have $(a, b)(c, d) = (0, 0)$ or $(c, d)(a, b) = (0, 0)$. $\text{Cay}(\mathbb{R}, -1)$ is commutative and therefore we only need to check either case. The product is the same as in equation 3.3, but with different conditions.

$$\begin{cases} 0 = ac - db = a^2c - adb \\ 0 = da + bc = dab + b^2c. \end{cases}$$

We add these above equations together and obtain $0 = a^2c + b^2c = c(a^2 + b^2)$. Now, if $a^2 + b^2 = 0$ we have $a = b = 0$ and hence (a, b) is not a zero divisor since it is not nonzero. Therefore $c = 0$, which means $db = 0$ and $da = 0$. If $d \neq 0 \implies a = b = 0$ and therefore $d = 0$. But this means $(c, d) = (0, 0)$, which is also not nonzero. Since these are the only possibilities, our assumption must be wrong and there are no zero divisors.

We have proven that $\text{Cay}(\mathbb{R}, -1)$ only has units and no zero divisors and thus $\text{Cay}(\mathbb{R}, -1)$ is a division algebra. This concludes the proof. $\square$

Armed with the knowledge that we obtain the complex numbers through the Cayley–Dickson construction by $\text{Cay}(\mathbb{R}, -1)$, we shall now discuss the quaternions in more detail.

3.1.2 The Quaternions $\mathbb{H}$

The quaternions, $\mathbb{H}$, is a vector space over $\mathbb{R}$ with basis $\mathcal{B} = \{1, i, j, k\}$. Each element in $\mathbb{H}$ can be written as $h = \lambda_1 1 + \lambda_2 i + \lambda_3 j + \lambda_4 k$ where $\lambda_i \in \mathbb{R}$ for $i = 1, 2, 3, 4$. If h_1 is yet another quaternion and if we define a product between quaternions as $hh_1 = (\lambda_1 1 + \lambda_2 i + \lambda_3 j + \lambda_4 k)(\lambda_a 1 + \lambda_b i + \lambda_c j + \lambda_d k) = \lambda_m 1 + \lambda_n i + \lambda_o j + \lambda_p k$ subject to $i^2 = j^2 = k^2 = ijk = -1$. Then, $\mathbb{H}$ is an algebra. The quaternion conjugation is $h^* = \lambda_1 1 - \lambda_2 i - \lambda_3 j - \lambda_4 k$. Furthermore, the quaternions can be seen as a vector space over $\mathbb{C}$ where the basis is $\mathcal{B}_C = \{1, j\}$ and a quaternion is then written as $h = c_1 1 + c_2 j$ where $c_1, c_2 \in \mathbb{C}$. This vector space is $\mathbb{C} \oplus \mathbb{C}$. The conjugation for such a quaternion is $h^* = c_1^* + c_2^* j$ where c_1^*, c_2^* is the ordinary conjugation for complex numbers and where $ij = k$. Like the complex numbers, the quaternions are not an ordered field. However, unlike the real numbers and the complex numbers, the quaternions are not commutative.

Proposition 12. *$\text{Cay}(\mathbb{C}, -1)$ is not a commutative algebra.*

Proof. Let $(i, 0), (0, 1) \in \text{Cay}(\mathbb{C}, -1)$. Then, by the commutator, we have

$$\begin{aligned} [(i, 0), (0, 1)] &= (i, 0)(0, 1) - (0, 1)(i, 0) = \\ &= (i \cdot 0 - 1 \cdot 0, 1 \cdot i + 0) - (0 \cdot i - 0 \cdot 1, 0 + 1 \cdot i^*) = \\ &= (0, i) - (0, -i) = (0, 2i) \neq (0, 0). \end{aligned}$$

Since commutativity does not hold for the elements $(i, 0)$ and $(0, 1)$, $\text{Cay}(\mathbb{C}, -1)$ is not commutative. This completes the proof. $\square$

Even though the quaternions are not commutative, we can prove that the quaternions are associative.

Proposition 13. *$\text{Cay}(\mathbb{C}, -1)$ is an associative algebra.*

Proof. Let $(a, b), (c, d), (e, f) \in \text{Cay}(\mathbb{C}, -1)$ where $a, b, \dots, f \in \text{Cay}(\mathbb{R}, -1)$. Then, by using the associator, we have

$$\begin{aligned} [(a, b), (c, d), (e, f)] &= ((a, b)(c, d))(e, f) - (a, b)((c, d)(e, f)) = \\ &= (ac - d^*b, da + bc^*)(e, f) - (a, b)(ce - f^*d, fc + de^*) = \\ &= ((ac - d^*b)e - f^*(da + bc^*), f(ac - d^*b) + (da + bc^*)e^*) \\ &\quad - (a(ce - f^*d) - (fc + de^*)^*b, (fc + de^*)a + b(ce - f^*d)^*) = \\ &= (ace - d^*be - f^*da - f^*bc^*, fac - fd^*b + dae^* + bc^*e^*) \\ &\quad - (ace - af^*d - (c^*f^* + ed^*)b, fca + de^*a + b(e^*c^* - d^*f)) = \\ &= (ace - d^*be - f^*da - f^*bc^*, fac - fd^*b + dae^* + bc^*e^*) - \\ &= (ace - af^*d - c^*f^*d - ed^*b, fca + de^*a + be^*c^* - bd^*f) = (0, 0) \implies \\ &= ((a, b)(c, d))(e, f) = (a, b)((c, d)(e, f)). \end{aligned}$$

We have used that the complex numbers are both commutative and associative. Since these are arbitrary quaternions and the statement holds, the quaternions are associative. This concludes the proof. $\square$

Proposition 14. $\text{Cay}_-(\mathbb{C}, -1) \cong^* \mathbb{H}$.

Proof. Let $\phi : \text{Cay}(\mathbb{C}, -1) \rightarrow \mathbb{H}$ be the mapping defined by $(a, b) \mapsto a + bj = a_1 + a_2i + b_1j + b_2k$ where $a := (a_1, a_2), b := (b_1, b_2) \in \text{Cay}(\mathbb{C}, -1)$. We ought to prove that ϕ is a $*$ -algebra isomorphism. First, we show that ϕ is $\mathbb{R}$ -linear. Let $(a, b), (c, d) \in \text{Cay}(\mathbb{C}, -1)$ and $\alpha, \beta \in \mathbb{R}$. Then, $\phi(\alpha(a, b) + \beta(c, d)) = \phi((\alpha a + \beta c, \alpha b + \beta d)) = (\alpha a + \beta c) + (\alpha b + \beta d)j = \alpha(a + bj) + \beta(c + dj) = \alpha\phi((a, b)) + \beta\phi((c, d))$. Hence, ϕ is a $\mathbb{R}$ -linear mapping. Furthermore, we have $\phi((a, b)(c, d)) = \phi((ac - d^*b, da + bc^*)) = (ac - d^*b) + (da + bc^*)j$. Furthermore, we have $\phi((a, b))\phi((c, d)) = (a + bj)(c + dj) = (ac + bjdj) + (adj + bjc) = (ac + bj^2d^*) + (adj + bc^*j) = (ac - d^*b) + (ad + bc^*)j$. Thus, $\phi((a, b)(c, d)) = \phi((a, b))\phi((c, d))$. Hereby, ϕ is a homomorphism. We can show that this mapping is injective. $\phi((a, b)) = \phi((c, d)) \iff a + bj = c + dj$, hence $a = c$ and $b = d$, which means $(a, b) = (c, d)$. Surjectivity is immediate. Lastly, we have $\phi((a, b)^*) = \phi((a^*, -b)) = a^* - bj = a_1 - a_2i - b_1j - b_2k = (a_1 + a_2i + b_1j + b_2k)^* = (a + bj)^* = \phi((a, b))^*$. Therefore it follows that ϕ is a $*$ -algebra isomorphism and hence $\text{Cay}(\mathbb{C}, -1) \cong^* \mathbb{H}$. $\square$

Definition 62. Let (a, b, c, d) be an ordered pair where $a, b, c, d \in \mathbb{R}$. Then, the imaginary units in $\text{Cay}(\mathbb{C}, -1)$ are represented by the following in $\mathbb{H} = \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R}$ over $\mathbb{R}$:

i is represented by $= (0, 1, 0, 0)$

j is represented by $= (0, 0, 1, 0)$

k is represented by $= (0, 0, 0, 1)$.

Furthermore, in the basis $\mathcal{B}_C = \{1, j\}$, we have the following instead.

Definition 63. Let (a, b) be an ordered pair where $a, b \in \mathbb{C}$. Then, the imaginary units in $\text{Cay}(\mathbb{C}, -1)$ are represented by the following in $\mathbb{C} \oplus \mathbb{C}$ over $\mathbb{C}$:

i is represented by $= (i, 0)$

j is represented by $= (0, 1)$

k is represented by $= (0, i)$.

Proposition 15 ($\text{Cay}(\mathbb{C}, -1)$ is an inner product space). Define a bilinear form $\langle \cdot | \cdot \rangle : \text{Cay}(\mathbb{C}, -1) \rightarrow \mathbb{R}$ by $\langle a | b \rangle = \frac{1}{2}(a^*b + b^*a)$. Then, this mapping is a norm and $\text{Cay}(\mathbb{C}, -1)$ is an inner product space.

Proof. The proof is similar to the proof for $\text{Cay}(\mathbb{R}, -1)$ and hence omitted. $\square$

The standard norm on $\mathbb{H}$ is $\|h\| = \sqrt{hh^*} = \sqrt{(a + bi + cj + dk)(a - bi - bj - bk)} = \sqrt{a^2 + b^2 + c^2 + d^2}$. Since $\phi : \text{Cay}_-(\mathbb{C}, -1) \rightarrow \mathbb{H}$ is a $*$ -algebra isomorphism, we have that $\phi^{-1} : \mathbb{H} \rightarrow \text{Cay}_-(\mathbb{C}, -1)$ is a $*$ -algebra isomorphism. We can therefore show that we get exactly the same norm with our ordered pair through the Cayley–Dickson construction.

Proposition 16 (A norm on $\text{Cay}(\mathbb{C}, -1)$). If $\|\cdot\|$ is a norm on $\mathbb{H}$ and $\phi : \text{Cay}(\mathbb{C}, -1) \rightarrow \mathbb{H}$ is a $*$ -algebra isomorphism, we have that $\phi^{-1}(\|\cdot\|)$ is a norm on $\text{Cay}(\mathbb{C}, -1)$.

Proof. First let $h = a + bi + cj + dk \in \mathbb{H}$. Then, we have $\phi^{-1}(\|h\|) = \phi^{-1}(\sqrt{a^2 + b^2 + c^2 + d^2}) = (\sqrt{a^2 + b^2 + c^2 + d^2}, 0, 0, 0)$. $\square$

The quaternions is an absolute valued division algebra. However, we will not show this.

3.1.3 The Octonions $\mathbb{O}$

The octonions form an 8-dimensional vector space over $\mathbb{R}$ with basis $\mathcal{B} = \{e_0, e_1, \dots, e_7\}$. Each element in $\mathbb{O}$ can be written as $o = \lambda_0 e_0 + \lambda_1 e_1 + \dots + \lambda_7 e_7$ where $\lambda_i \in \mathbb{R}$ for $i = 1, 2, \dots, 7$. Multiplication between basis vectors can be seen in Table 3.1, below.

	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	$-e_0$	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	$-e_0$	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	$-e_0$	e_1	e_2	e_3
e_5	e_5	e_4	$-e_7$	e_6	$-e_1$	$-e_0$	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	$-e_0$	$-e_1$
e_7	e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	$-e_0$

Table 3.1: Multiplication table for $\mathbb{O}$ as seen as a vector space over $\mathbb{R}$.

Proposition 17. $\text{Cay}_-(\mathbb{H}, -1)^* \cong \mathbb{O}$.

Proof. The proof is similar to those in Sections 3.1.1 and 3.1.2 and is therefore omitted. $\square$

In the multiplication Table 3.1 one may observe that the octonions are neither commutative nor associative. Instead, the octonions have the property of being alternative. We prove this next.

Proposition 18. $\text{Cay}(\mathbb{H}, -1)$ is an alternative algebra.

Proof. Let $(a, b), (c, d) \in \text{Cay}(\mathbb{H}, -1)$ where $a, b, c, d \in \text{Cay}(\mathbb{C}, -1)$. To show that the octonions are alternative, we must show that $(a, b)((a, b)(c, d)) = ((a, b)(a, b))(c, d)$ and $((c, d)(a, b))(a, b) = (c, d)((a, b)(a, b))$.

$$\begin{aligned}
 (a, b)((a, b)(c, d)) &= (a, b)(ac - d^*b, da + bc^*) \\
 &= (a(ac - d^*b) - (da + bc^*)^*b, (da + bc^*)a + b(ac - d^*b)^*) \\
 &= (aac - ad^*b - a^*d^*b - cb^*b, daa + bc^*a + bc^*a^* - bb^*d)
 \end{aligned}$$

$$\begin{aligned}
 ((a, b)(a, b))(c, d) &= (aa - b^*b, ba + ba^*)(c, d) \\
 &= ((aa - b^*b)c - d^*(ba + ba^*), d(aa - b^*b) + (ba + ba^*)c^*) \\
 &= (aac - b^*bc - d^*ba - d^*ba^*, daa - db^*b + bac^* + ba^*c^*)
 \end{aligned}$$

Now, we need the following to be equal:

$$\begin{aligned}
& \begin{cases} aac - ad^*b - a^*d^*b - cb^*b = aac - b^*bc - d^*ba - d^*ba^* \\ daa + bc^*a + bc^*a^* - bb^*d = daa - db^*b + bac^* + ba^*c^* \end{cases} \iff \\
& \begin{cases} -ad^*b - a^*d^*b - cb^*b = -b^*bc - d^*ba - d^*ba^* \\ bc^*a + bc^*a^* - bb^*d = db^*b + bac^* + ba^*c^* \end{cases} \iff \\
& \begin{cases} -(a + a^*)d^*b - cb^*b = -b^*bc - d^*b(a + a^*) \\ bc^*(a + a^*) - bb^*d = db^*b + b(a + a^*)c^* \end{cases}
\end{aligned}$$

Since b^*b is a real number, it commutes with all other quaternions and hence $cb^*b = b^*bc$. Furthermore, $a + a^*$ is also a real number so it commutes with all quaternions, hence both equations above are the same and we have $(a, b)((a, b)(c, d)) = ((a, b)(a, b))(c, d)$. The proof for $((c, d)(a, b))(a, b) = (c, d)((a, b)(a, b))$ is similar and omitted. Hereby we can conclude that the octonions is an alternative algebra. This concludes the proof. $\square$

A natural norm and inner product is similar to the norm for the complex numbers and the quaternions in sections 3.1.1 and 3.1.2, although we will not show this. Similar to the complex numbers and the quaternions, the octonions form an absolute valued division algebra [3].

3.2 The Standard Norm for the Cayley–Dickson Algebras

By the previous section, we already have an idea for a natural norm for the algebras $Cay(\mathbb{R}, -1)$, $Cay(\mathbb{C}, -1)$ and $Cay(\mathbb{H}, -1)$ over $\mathbb{R}$ with the identity mapping on $\mathbb{R}$ and $\mu = -1$. Let $C = \{Cay(\mathbb{R}, -1), Cay(\mathbb{C}, -1), Cay(\mathbb{H}, -1)\}$ and by $A \in C$, we mean any algebra in C .

Proposition 19 (Standard inner product). $\langle a|b \rangle = \frac{1}{2}(a^*b + b^*a, 0)$ where $a, b \in A$ for any $A \in C$ is an inner product on any algebra $A \in C$. We call this the standard norm.

Proof. Proof is similar to the one for complex numbers in Section 3.1.1, as that is a special case for the general case. Therefore the proof is omitted. $\square$

Lemma 4. The standard inner product for the algebras in C satisfies $\langle a|c \rangle + \langle b|d \rangle = \langle (a, b)|(c, d) \rangle$.

Proof. The proof is straight forward by computation. Let $a, b, c, d \in A$. Then we have

$$\begin{aligned}
\langle (a, b)|(c, d) \rangle &= \frac{1}{2} \left(((a, b)^*(c, d) + (c, d)^*(a, b)) \right) = \frac{1}{2} \left(((a^*, -b)(c, d) + (c^*, -d)(a, b)) \right) \\
&= \frac{1}{2} \left((a^*c + d^*b, da^* - bc^*) + (c^*a + b^*d, bc^* - da^*) \right) \\
&= \frac{1}{2} \left((a^*c + c^*a + b^*d + d^*b, da^* - bc^* + bc^* - da^*) \right) \\
&= \frac{1}{2} (a^*c + c^*a + b^*d + d^*b, 0) = \frac{1}{2} (a^*c + c^*a, 0) + \frac{1}{2} (b^*d + d^*b, 0) = \langle a|c \rangle + \langle b|d \rangle.
\end{aligned}$$

This completes the proof. $\square$

Proposition 20 (The standard norm). $\|(a, b)\| = \sqrt{(a, b)(a, b)^*}$ is a norm for any algebra $A \in \mathcal{C}$. We call this the standard norm.

Proof. Let A be a $*$ -algebra. Then we have $(a, b), (a, b)^* \in A \oplus A$, which if we multiply gives

$$(a, b)(a, b)^* = \sqrt{(aa^* + b^*b, -ba + ba)} = \sqrt{(aa^* + b^*b, 0)} = \sqrt{aa^* + b^*b}.$$

We note that both aa^* and bb^* are real numbers and hence this product maps to $\mathbb{R}$. Thus, we obtain $\|(a, b)\| = \sqrt{(a, b)(a, b)^*} = \sqrt{aa^* + b^*b}$. This is only 0 if $(a, b) := (0, 0)$ and otherwise strictly positive. Hence, the first requirement in Definition 35 is satisfied. For the second requirement, first let $\alpha \in \mathbb{R}$. Then we have

$$\|\alpha(a, b)\| = \sqrt{\alpha^2(aa^* + b^*b, 0)} = \sqrt{\alpha^2} \sqrt{(aa^* + b^*b, 0)} = |\alpha| \|(a, b)\|.$$

Where we used the absolute value for α as it is real. Therefore, the second requirement is satisfied. Lastly, we have the so-called triangle inequality.

$$\begin{aligned} \|(a, b) + (c, d)\|^2 &= \|(a + c, b + d)\|^2 = (aa^* + b^*b + cc^* + d^*d + ac^* + ca^* + b^*d + d^*b, 0) \\ &= (aa^* + b^*b, 0) + (cc^* + d^*d, 0) + (ac^* + ca^* + b^*d + d^*b, 0) \\ &= \|(a, b)\|^2 + \|(c, d)\|^2 + 2\langle a|c \rangle + 2\langle b|d \rangle \\ &= \|(a, b)\|^2 + \|(c, d)\|^2 + 2\langle (a, b)|(c, d) \rangle \leq \|(a, b)\|^2 + \|(c, d)\|^2 + 2\|(a, b)\|\|(c, d)\| \\ &\implies \|(a, b) + (c, d)\| \leq \|(a, b)\| + \|(c, d)\| \end{aligned}$$

Where we have utilized the Cauchy Schwarz inequality to prove the triangle inequality. This completes the proof. $\square$

3.3 The Split Versions

We will mention the split versions briefly as our new algebras later might be isomorphic to one of these. We create the split-complexes by constructing $\text{Cay}(\mathbb{R}, 1)$. Thereafter we can put the split-complexes into the construction and obtain the split-quaternions and similar for the split-octonions.

3.3.1 The Split Complexes $\mathbb{C}'$

The split-complexes, $\mathbb{C}'$, is a vector space over $\mathbb{R}$ with basis $\mathcal{B} = \{1, i\}$. Each element in $\mathbb{C}'$ can be written as $c = \lambda_1 1 + \lambda_2 i$ where $\lambda_i \in \mathbb{R}$ for $i = 1, 2$. The split-complexes conjugation is $c^* = \lambda_1 1 - \lambda_2 i$. One can show that $\text{Cay}(\mathbb{R}, 1) \overset{*}{\cong} \mathbb{C}'$ through the mapping $(a, b) = a + bi$.

	(0, 1)	(0, 1)
(1, 0)	(1, 0)	(0, 1)
(0, 1)	(0, 1)	(1, 0)

Table 3.2: Multiplication table for $\text{Cay}(\mathbb{R}, 1)$ as seen as a vector space over $\mathbb{R}$.

	1	i
1	1	i
i	i	1

Table 3.3: Multiplication table for $\mathbb{C}'$ as seen as a vector space over $\mathbb{R}$.

We see that the imaginary unit, i , satisfies the relationship $i^2 = 1$.

3.3.2 The Split Quaternions $\mathbb{H}'$

The split-quaternions, $\mathbb{H}'$, is a vector space over $\mathbb{R}$ with basis $\mathcal{B} = \{1, i, j, k\}$. Each element in $\mathbb{H}'$ can be written as $h = \lambda_1 1 + \lambda_2 i + \lambda_3 j + \lambda_4 k$ where $\lambda_i \in \mathbb{R}$ for $i = 1, 2, 3, 4$. The split-quaternion conjugation is $h^* = \lambda_1 1 - \lambda_2 i - \lambda_3 j - \lambda_4 k$. Furthermore, the split-quaternions can be seen as a vector space over $\mathbb{C}'$ where the basis is $\mathcal{B}_C = \{1, j\}$ and a split-quaternion is then written as $h = c_1 a + c_2 j$ where $c_1, c_2 \in \mathbb{C}'$. This vector space is $\mathbb{C}' \oplus \mathbb{C}'$. The conjugation for such a split-quaternion is $h^* = c_1^* + c_2^* j$ where c_1^*, c_2^* is the ordinary conjugation for split-complexes. The split-quaternions are not commutative but are associative, however we will not show this. Furthermore, the split-quaternions is neither a normed algebra nor a division algebra. We show the multiplication table for the split-quaternions, below.

	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	1	$-i$
k	k	j	i	1

Table 3.4: Multiplication table for $\mathbb{H}'$ as seen as a vector space over $\mathbb{R}$.

The following relationship between the basis vectors hold

$$\begin{cases} i^2 = -1 \\ j^2 = k^2 = 1 \\ ij = k \\ ik = -j \\ jk = -i. \end{cases}$$

One can show that the split-quaternions can be obtained by constructing $\text{Cay}(\mathbb{C}', 1)$, where $\mathbb{C}'$ are the split complex numbers and by using the standard involution on $\text{Cay}(\mathbb{C}', 1)$. Therefore, without proof, $\text{Cay}(\mathbb{C}', 1) \stackrel{*}{\cong} \mathbb{H}'$.

3.3.3 The Split Octonions $\mathbb{O}'$

The Split octonions form an 8-dimensional vector space over $\mathbb{R}$ with basis $\mathcal{B} = \{e_0, e_1, \dots, e_7\}$. Each element in $\mathbb{O}'$ can be written as $o = \sum_{i=0}^7 \lambda_i e_i$ where $\lambda_i \in \mathbb{R}$ for $i = 1, 2, \dots, 7$. The split-

octonion conjugation is $o^* = \lambda_0 e_0 - \sum_{i=1}^7 \lambda_i e_i$. Furthermore, one can show that $\text{Cay}(\mathbb{H}', 1) \stackrel{*}{\cong} \mathbb{C}'$ by the mapping $\phi : \text{Cay}_+(\mathbb{H}', 1) \rightarrow \mathbb{O}'$ through the mapping $\phi((a, b)) = a + bl$. We show multiplication table for the the split-octonions, below.

	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	e_0	$-e_1$	e_6	e_7	e_4	e_5
e_3	e_3	e_2	e_1	e_0	e_7	$-e_6$	$-e_5$	e_4
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	e_0	$-e_1$	$-e_2$	$-e_3$
e_5	e_5	e_4	$-e_7$	e_6	e_1	e_0	e_3	$-e_2$
e_6	e_6	e_7	$-e_4$	e_5	e_2	$-e_3$	$-e_0$	$-e_1$
e_7	e_7	$-e_6$	$-e_5$	$-e_4$	e_3	e_2	e_1	$-e_0$

Table 3.5: Multiplication table for $\mathbb{O}'$ as seen as a vector space over $\mathbb{R}$.

Chapter 4

A Novel Involution and its Induced Algebras

4.1 A Novel Involution

The standard involution maps b to $-b$. A different natural choice would be to map b to b . We prove that this is indeed an involution on $\text{Cay}(A, \mu)$.

Proposition 21 (A different involution). *Let A be a $*$ -algebra and define a mapping $*$: $A \oplus A \rightarrow A \oplus A$ by $(a, b)^* = (a^*, b)$. Then $*$ is an involution on $\text{Cay}(A, \mu)$.*

Proof. We first prove that the first part in definition 30 holds. Let $(a, b) \in A \oplus A$. Then, we have

$$((a, b)^*)^* = (a^*, b)^* = ((a^*)^*, b) = (a, b).$$

Hence the first requirement holds. Now for the second part, we let $(a, b), (c, d) \in A \oplus A$.

$$\begin{aligned} ((a, b)(c, d))^* &= (ac + \mu d^* b, da + bc^*)^* = ((ac + \mu d^* b)^*, da + bc^*) \\ &= ((ac)^* + (\mu d^* b)^*, da + bc^*) = (c^* a^* + b^* d \mu^*, da + bc^*) = (c^* a^* + \mu b^* d, da + bc^*). \\ (c, d)^*(a, b)^* &= (c^*, d)(a^*, b) = (c^* a^* + \mu b^* d, bc^* d(a^*)^*) \\ &= (c^* a^* + \mu b^* d, da + bc^*). \end{aligned}$$

Therefore the second part also holds. We must also show that this mapping is F -linear. First, let $\alpha, \beta \in F$.

$$\begin{aligned} (\alpha(a, b) + \beta(c, d))^* &= (\alpha a + \beta c, \alpha b + \beta d)^* = ((\alpha a + \beta c)^*, \alpha b + \beta d) = \\ &= (\alpha a^* + \beta c^*, \alpha b + \beta d) = (\alpha a^*, \alpha b) + (\beta c^*, \beta d) = \alpha(a^*, b) + \beta(c^*, d) = \alpha(a, b)^* + \beta(c, d)^*. \end{aligned}$$

Hence, the mapping is F -linear and we have proven this is an involution. $\square$

4.2 Starting Algebra is $\mathbb{R}$

We will now utilize the involution given in Proposition 21. We denote this different conjugation with sub scripted plus sign as such: $Cay_+(\mathbb{R}, -1)$, where we start with $\mathbb{R}$ and the parameter $\mu = -1$. On the algebra $\mathbb{R}$ we obly have the identity involution $a^* = a$.

4.2.1 $Cay_+(\mathbb{R}, -1)$

If we multiply two elements in $Cay_+(\mathbb{R}, -1)$ we get

$$(a, b)(c, d) = (ac - d^*b, da + bc^*) = (ac - db, da + bc). \quad (4.1)$$

Thus, we have that $Cay(\mathbb{R}, -1) \cong \mathbb{C}$, the proof is similar to Proposition 7. Yet, our involution is different this time. Therefore we know that $Cay_+(\mathbb{R}, -1) \not\cong \mathbb{C}$. The conjugation for any $(a, b) \in Cay_+(\mathbb{R}, -1)$ is $(a, b)^* = (a, b)$. This means that $(a, b)^* = (a, b)$. If we multiply any member in $\mathbb{R} \oplus \mathbb{R}$ with its conjugate, we obtain

$$(a, b)(a, b)^* = (a^2 - b^2, 2ba).$$

Through the above computation, we can observe that in general we do not get a non-negative real number and hence a norm can not be defined through the conjugation in a natural way as with the previous involution. Since $Cay_+(\mathbb{R}, -1) = \mathbb{C}$, the algebra is associative and commutative but have no order. The proofs are omitted since the proofs are the same as in section 3.1.1. Furthermore, $(a, b)(a, b)^* \in \mathbb{R}$ if and only if $2ba = 0 \implies a = 0$ or $b = 0$. Ultimately, since we have $a^2 - b^2 \geq 0 \implies a^2 \geq b^2$. By combining both criteria, we see that $b = 0$ is the only possible value that satisfies both restrictions. Hence the algebra $A \subset Cay_+(\mathbb{R}, -1)$ that has a norm defined by the product of an element and its conjugation is the algebra $A = \{(a, b) : b = 0\}$. We observe that the algebra A is the real numbers, $\mathbb{R}$.

4.2.2 $Cay_+(\mathbb{C}, -1)$

We continue the construction with $Cay_+(\mathbb{C}, -1)$ and the same involution $(a, b)^* = (a^*, b)$. The space is $\mathbb{C} \oplus \mathbb{C}$, furthermore since conjugation for the complex numbers leaves everything unchanged. We have already shown that this algebra is commutative and associative.

We may examine the multiplication between an element in $Cay_+(\mathbb{C}, -1)$ and its conjugate. Therefore, let $(a, b) \in Cay_+(\mathbb{C}, -1)$.

$$(a, b)(a, b)^* = (a, b)(a^*, b) = (a, b)(a, b) = (aa - bb, ba + ba). \quad (4.2)$$

We know that a complex number (a, b) where $a, b \in \mathbb{R}$ gave $(a, b)(a, b)^* = (a^2 - b^2, 2ba)$. Therefore, in terms of real numbers and if we let $a := (a, b)$ and $b := (c, d)$ where $a, b, c, d \in \mathbb{R}$, equation 4.2 becomes:

$$\begin{aligned} (aa - bb, 2ba) &= ((a^2 - b^2, 2ba) - (c^2 - d^2, 2cd), 2(ac - db, da + bc)) \\ &= ((a^2 - b^2 - c^2 + d^2, 2ba - 2cd), (2ac - 2db, 2da + 2bc)). \end{aligned}$$

If the conjugation ought to define a norm, we must have the criteria:

$$\begin{cases} a^2 - b^2 - c^2 + d^2 \geq 0 \\ ba - cd = 0 \\ ac - db = 0 \\ da + bc = 0 \end{cases}$$

It is evident that since we started with $\mathbb{R}$ and the identity mapping, commutativity and associativity remained in each doubling step. Therefore, we try a different starting algebra instead.

4.3 Starting Algebra is $\mathbb{C}$

We start with $\mathbb{C}$ and the involution $*$: $(a, b) \mapsto (a, -b)$ on $\mathbb{C}$. This is the ordinary complex conjugation. Thereafter, we construct the algebra $\text{Cay}_+(\mathbb{C}, -1) = \mathcal{H}$.

4.3.1 $\text{Cay}_+(\mathbb{C}, -1)$

For $\text{Cay}_+(\mathbb{C}, -1)$, the vector space is $\mathbb{C} \oplus \mathbb{C}$. Therefore each vector in $\mathbb{C} \oplus \mathbb{C}$ can be written as $h = c_1 + c_2j$, where $c_1, c_2 \in \mathbb{C}$. Hence a basis for this vector space is $\mathcal{B}_C = \{1, j\}$. We observe that the corresponding basis vectors in $\text{Cay}_+(\mathbb{C}, -1)$ would be $(1, 0)$, $(i, 0)$, $(0, 1)$ and $(0, i)$ where $0, 1, i \in \mathbb{C}$. This algebra seems similar to $\mathbb{H}$ and in fact we have already shown $\text{Cay}_-(\mathbb{C}, -1) \cong^* \mathbb{H}$ and hence $\text{Cay}(\mathbb{C}, -1) \cong \mathbb{H}$. Therefore, commutativity does not hold but associativity does.

Note that we only described $\text{Cay}_+(\mathbb{C}, -1)$ and $\mathbb{H}$ through an isomorphism and not a $*$ -algebra isomorphism. We prove this next.

Lemma 5. $\text{Cay}_+(\mathbb{C}, -1) \not\cong^* \mathbb{H}$.

Proof. We must show that $\phi((a, b)^*) \neq \phi(a, b)^*$, where ϕ is the mapping in Proposition 14.

$$\begin{aligned} \phi((a, b)^*) &= \phi((a^*, b)) = a^* + bj = a_1 - a_2i + b_1j + b_2k \\ \phi((a, b))^* &= (a + bj)^* = (a_1 + a_2i + b_1j + b_2k)^* = a_1 - a_2i - b_1j - b_2k. \end{aligned}$$

Evidently, $\phi((a, b)^*) \neq \phi((a, b))^*$ and ultimately $\text{Cay}_+(\mathbb{C}, -1) \not\cong^* \mathbb{H}$. This completes the proof. $\square$

We show the multiplication for the algebra $\mathcal{H}$ both in terms of complex numbers and of the ordinary basis for the quaternions. This can be done since $\text{Cay}(\mathbb{C}, -1) \cong \mathbb{H}$.

	$(1, 0)$	$(i, 0)$	$(0, 1)$	$(0, i)$
$(1, 0)$	$(1, 0)$	$(i, 0)$	$(0, 1)$	$(0, i)$
$(i, 0)$	$(i, 0)$	$-(1, 0)$	$(0, i)$	$-(0, 1)$
$(0, 1)$	$(0, 1)$	$-(0, i)$	$-(1, 0)$	$(i, 0)$
$(0, i)$	$(0, i)$	$(0, 1)$	$-(i, 0)$	$-(1, 0)$

Table 4.1: Multiplication table for $\text{Cay}(\mathbb{C}, -1)$ as seen as a vector space over $\mathbb{R}$.

	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

Table 4.2: Multiplication table for $\mathbb{H}$ as seen as a vector space over $\mathbb{R}$.

Since $\text{Cay}(\mathbb{C}, -1) \cong \mathbb{H}$ the multiplication for $\text{Cay}_+(\mathbb{C}, -1)$ is the same as for $\mathbb{H}$ and the ordinary relationship between the basis vectors hold, as seen as an algebra over $\mathbb{R}$. Since conjugation only flips the sign for the basis vector i , the algebra we would obtain by doubling $\text{Cay}(\mathcal{H}, -1)$ would be 8-dimensional but should not obey the multiplication table given by the octonions in Table 3.1. We should note that the basis vectors under conjugation gives

$$\left\{ \begin{array}{l} 1^* = 1 \in \mathbb{H} \text{ is represented by } (1, 0)^* = (1^*, 0) = (1, 0) \in \text{Cay}_+(\mathbb{C}, -1), \\ i^* = -i \in \mathbb{H} \text{ is represented by } (i, 0)^* = (i^*, 0) = (-i, 0) \in \text{Cay}_+(\mathbb{C}, -1), \\ j^* = j \in \mathbb{H} \text{ is represented by } (0, 1)^* = (0, 1) = (0, 1) \in \text{Cay}_+(\mathbb{C}, -1), \\ k^* = k \in \mathbb{H} \text{ is represented by } (0, i)^* = (0, i) = (0, i) \in \text{Cay}_+(\mathbb{C}, -1). \end{array} \right.$$

4.3.2 $\text{Cay}_+(\mathcal{H}, -1)$

We continue the doubling of algebras to obtain the vector space $\mathcal{H} \oplus \mathcal{H}$ as seen over $\mathcal{H}$ or the vector space $\bigoplus_{i=1}^8 \mathbb{R}$ over $\mathbb{R}$. This is an 8-dimensional vector space over $\mathbb{R}$. Hence a basis is $\mathcal{B} = \{e_0, e_1, \dots, e_7\}$. A basis for $\mathcal{H} \oplus \mathcal{H}$ as seen over $\mathcal{H}$ would be $\mathcal{B}_H = \{1, l\}$ where each element in $\text{Cay}_+(\mathcal{H}, -1) = \mathcal{O}$ would be written as $o = a_1 + a_2 l$ and where $a_1, a_2 \in \mathcal{H}$. Since the involution on $\mathcal{H} \oplus \mathcal{H}$ is different from the involution given by the quaternions. The basis $\mathcal{B}_H = \{1, l\}$ should yield multiplication between basis vectors different than for those in the octonions. Furthermore, we observe that the basis vectors $e_i = (a_1, a_2)$ where $a_1, a_2 \in \mathcal{H}$ and $i = 0, 1, \dots, 7$, can be written as

$$\begin{array}{ll} e_0 \in \mathcal{O} \text{ is represented by } (1, 0) \in \text{Cay}(\mathcal{H}, -1) & e_1 \in \mathcal{O} \text{ is represented by } (i, 0) \in \text{Cay}(\mathcal{H}, -1) \\ e_2 \in \mathcal{O} \text{ is represented by } (j, 0) \in \text{Cay}(\mathcal{H}, -1) & e_3 \in \mathcal{O} \text{ is represented by } (k, 0) \in \text{Cay}(\mathcal{H}, -1) \\ e_4 \in \mathcal{O} \text{ is represented by } (0, 1) \in \text{Cay}(\mathcal{H}, -1) & e_5 \in \mathcal{O} \text{ is represented by } (0, i) \in \text{Cay}(\mathcal{H}, -1) \\ e_6 \in \mathcal{O} \text{ is represented by } (0, j) \in \text{Cay}(\mathcal{H}, -1) & e_7 \in \mathcal{O} \text{ is represented by } (0, k) \in \text{Cay}(\mathcal{H}, -1) \end{array}$$

We present the multiplication tables for the algebra $\text{Cay}(\mathcal{H}, -1)$, now.

	(1, 0)	(i, 0)	(j, 0)	(k, 0)	(0, 1)	(0, i)	(0, j)	(0, k)
(1, 0)	(1, 0)	(i, 0)	(j, 0)	(k, 0)	(0, 1)	(0, i)	(0, j)	(0, k)
(i, 0)	(i, 0)	-(1, 0)	(k, 0)	-(j, 0)	(0, i)	-(0, 1)	-(0, k)	(0, j)
(j, 0)	(j, 0)	-(k, 0)	-(1, 0)	(i, 0)	(0, j)	(0, k)	-(0, 1)	-(0, i)
(k, 0)	(k, 0)	(j, 0)	-(i, 0)	-(1, 0)	(0, k)	-(0, j)	(0, i)	-(0, 1)
(0, 1)	(0, 1)	-(0, i)	(0, j)	(0, k)	-(1, 0)	(i, 0)	-(j, 0)	-(k, 0)
(0, i)	(0, i)	(0, 1)	(0, k)	-(0, j)	-(i, 0)	-(1, 0)	(k, 0)	-(j, 0)
(0, j)	(0, j)	(0, k)	-(0, 1)	(0, i)	-(j, 0)	(k, 0)	(1, 0)	(i, 0)
(0, k)	(0, k)	-(0, j)	-(0, i)	-(0, 1)	-(k, 0)	-(j, 0)	-(i, 0)	(1, 0)

Table 4.3: Multiplication table for $\text{Cay}(\mathcal{H}, -1)$ as seen as a vector space over $\mathbb{R}$.

	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	$-e_0$	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	$-e_0$	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	e_6	e_7	$-e_0$	e_1	$-e_2$	$-e_3$
e_5	e_5	e_4	e_7	$-e_6$	$-e_1$	$-e_0$	e_3	$-e_2$
e_6	e_6	e_7	$-e_4$	e_5	$-e_2$	e_3	e_0	e_1
e_7	e_7	$-e_6$	$-e_5$	$-e_4$	$-e_3$	$-e_2$	$-e_1$	e_0

Table 4.4: Multiplication table for $\mathcal{O}$ as seen as a vector space over $\mathbb{R}$.

The green cells in Table 4.4 indicates a change compared to the basis vector multiplication of octonions in Table 3.1. Furthermore, each element in $\mathcal{O}$ can be written as $o = \sum_{i=0}^7 a_i e_i$ where $a_i \in \mathbb{R}$ $i = 1, 2, \dots, 7$. It follows from the involution on $\mathcal{H}$ that

$$o^* = a_0 e_0 - a_1 e_1 + \sum_{i=2}^7 a_i e_i.$$

We know that the only basis element that flips sign under conjugation is e_1 , e.g $e_1^* = -e_1$. Otherwise we have $e_i^* = e_i$ for $i = 0, 2, 3, \dots, 7$. As with the octonions, we suspect that $\text{Cay}(\mathcal{H}, -1)$ is not commutative, which we show first.

Proposition 22. *$\text{Cay}(\mathcal{H}, -1)$ is not a commutative algebra.*

Proof. Let $e_2, e_3 \in \text{Cay}(\mathcal{H}, -1)$. Then, by the commutator, we have

$$[e_2, e_3] = e_2 e_3 - e_3 e_2 = e_1 + e_1 = 2e_1 \neq 0.$$

Since the commutator is nonzero, $\text{Cay}(\mathcal{H}, -1)$ is not commutative. This completes the proof. $\square$

Furthermore, $\text{Cay}(\mathcal{H}, -1)$ is not associative.

Proposition 23. $\text{Cay}(\mathcal{H}, -1)$ is not an associative algebra.

Proof. Let $e_2, e_3, e_4 \in \text{Cay}(\mathcal{H}, -1)$. Then, by using the associator, we have

$$[e_2, e_3, e_4] = (e_2e_3)e_4 - e_2(e_3e_4) = e_1e_4 - e_2e_7 = e_5 + e_5 = 2e_5 \neq 0.$$

Since the associator is nonzero, $\text{Cay}(\mathcal{H}, -1)$ is not associative. This completes the proof. $\square$

Since the algebra seems to exhibit a different behavior compared to the octonions, we might suspect that it has zero divisors. The following definition is needed.

Definition 64 (Degree 2 zero divisor). Let $o \in \mathcal{O}$ where $o = a_0e_i + a_1e_j$ with $i \neq j$ and $i, j \in \{1, 2, \dots, 7\}$ and $a_0, a_1 \neq 0$. If there exists $o_1 \in \mathcal{O}$ where $o_1 = b_0e_i + b_1e_j$ with $i \neq j$ and $i, j \in \{1, 2, \dots, 7\}$ and $b_0, b_1 \neq 0$ such that o is a zero divisor, then we call o a *zero divisor of degree 2*.

Proposition 24. There exist left zero divisors of degree 2 in the algebra $\mathcal{O}$ over the field $\mathbb{R}$.

Proof. If $o_1 = b_0e_2 + b_1e_4$, then $o = a_0e_0 + a_1e_6$ is a left degree 2 zero divisor of o_1 .

$$\begin{aligned} oo_1 &= (a_0e_0 + a_1e_6)(b_0e_2 + b_1e_4) = a_0b_0e_2 + a_1b_0e_4 + a_1b_0e_6e_2 + a_1b_1e_6e_4 \\ &= (a_0b_0 - a_1b_1)e_2 + (a_0b_1 - a_1b_0)e_4. \end{aligned}$$

This is zero if $a_0b_0 - a_1b_1 = 0$ and $a_0b_1 - a_1b_0 = 0$. Since $a_0, a_1, b_0, b_1 \in \mathbb{R}$ we can find values nonzero such that the equations hold and equal zero. In fact these are $a_0 = a_1$ and $b_0^2 = b_1^2$. This completes the proof. $\square$

Corollary 1. There exist right zero divisors of degree 2 in the algebra $\mathcal{O}$ over the field $\mathbb{R}$.

Proof. The proof is immediate by performing the multiplication o_1o , where o and o_1 are as in Proposition 24. $\square$

We know that there are infinitely many left and right zero divisors of degree 2. Therefore, we extend our definition.

Definition 65 (Degree n zero divisor). Let $o \in \mathcal{O}$ where $o = \sum_{i=0}^7 a_ie_i$ and $a_0, a_1, \dots, a_7 \in \mathbb{R}$. Furthermore, let $o_1 \in \mathcal{O}$ where $o_1 = \sum_{i=0}^7 b_ie_i$ and $b_0, b_1, \dots, b_7 \in \mathbb{R}$. Then, if exactly n of $a_0, a_1, \dots, a_7$ are nonzero and if exactly n of $b_0, b_1, \dots, b_7$ are nonzero, we call o a *zero divisor of degree n* of o_1 if $oo_1 = 0$ or $o_1o = 0$.

Proposition 25. There exist left zero divisors of degree 4 in the algebra $\mathcal{O}$ over the field $\mathbb{R}$.

Proof. First, let $o = a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6$ and $o_1 = b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6$. Meaning $o, o_1 \in \mathcal{O}$. Then, we have

$$\begin{aligned} oo_1 &= (a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6)(b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6) = \\ &= (a_0b_0 - a_1b_1 - a_2b_2 + a_3b_3)e_0 + (a_0b_1 + a_1b_0 - a_2b_3 - a_3b_2)e_2 + \\ &= (a_0b_2 - a_1b_3 + a_2b_0 - a_3b_1)e_4 + (a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0)e_6 \end{aligned}$$

Furthermore, let $a_0 = a_1 = a_2 = -a_3 = b_0 = b_1 = -b_2 = b_3 \neq 0$. Then, we have

$$\begin{aligned} oo_1 &= (a_0e_0 + a_0e_2 + a_0e_4 - a_0e_6)(a_0e_0 + a_0e_2 - a_0e_4 + a_0e_6) = \\ &= (a_0^2 - a_0^2 + a_0^2 - a_0^2)e_0 + (a_0^2 + a_0^2 - a_0^2 - a_0^2)e_2 + \\ &+ (-a_0^2 - a_0^2 + a_0^2 + a_0^2)e_4 + (a_0^2 - a_0^2 + a_0^2 - a_0^2)e_6 = 0e_0 + 0e_2 + 0e_4 + 0e_6 = 0. \end{aligned}$$

Therefore o is a left zero divisor of degree 4 for o_1 . Since $a_0 \in \mathbb{R}$, there are infinitely many left zero divisors of degree 4. This completes the proof. $\square$

Corollary 2. *There exist right zero divisors of degree 4 in the algebra O over the field $\mathbb{R}$.*

Proof. The proof is immediate by performing the multiplication o_1o , where o and o_1 are as in Proposition 25. $\square$

By Propositions 24 and 25 and their respective corollaries, we propose the following to hold.

Conjecture 1 (Left degree n zero divisors of order 2^m). *There exist left zero divisors of degree n in the algebra O over the field $\mathbb{R}$ for any choice of $m \in \{1, 2, 3\}$ such that $2^m = n$.*

Since the basis vectors in Proposition 25 are closed under multiplication, we suspect that they form a subalgebra of O . We have that $O_{0246} = \text{span}(\mathcal{B}_{O_{0246}})$ over the field $\mathbb{R}$ where $\mathcal{B}_{O_{0246}} = \{e_0, e_2, e_4, e_6\}$ is a basis for O_{0246} . Therefore O_{0246} is a vector space.

Proposition 26. *O_{0246} is a subalgebra of O .*

Proof. We must show that O_{0246} is closed under multiplication. Therefore, let $o = a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6$ and $o_1 = b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6$ where $o, o_1 \in O_{0246}$.

$$\begin{aligned} oo_1 &= (a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6)(b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6) = \\ &= (a_0b_0 - a_1b_1 - a_2b_2 + a_3b_3)e_0 + (a_0b_1 + a_1b_0 - a_2b_3 - a_3b_2)e_2 + \\ &+ (a_0b_2 - a_1b_3 + a_2b_0 - a_3b_1)e_4 + (a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0)e_6 \end{aligned}$$

Hence, O_{0246} is a subalgebra of O . This completes the proof. $\square$

Theorem 5. *O_{0246} is a commutative algebra.*

Proof. We will show O_{0246} is a commutative algebra with the commutator. First, let $o = a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6$ and $o_1 = b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6$, e.g. $o, o_1 \in O_{0246}$. Then, we

have

$$\begin{aligned}
[o, o_1] &= oo_1 - o_1o = \left((a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6)(b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6) \right) - \\
&\left((b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6)(a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6) \right) = \\
&\left((a_0b_0 - a_1b_1 - a_2b_2 + a_3b_3)e_0 + (a_0b_1 + a_1b_0 - a_2b_3 - a_3b_2)e_2 + \right. \\
&(a_0b_2 - a_1b_3 + a_2b_0 - a_3b_1)e_4 + (a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0)e_6 \Big) \\
&- \left((a_0b_0 - a_1b_1 - a_2b_2 + a_3b_3)e_0 + (a_0b_1 + a_1b_0 - a_2b_3 - a_3b_2)e_2 + \right. \\
&(a_0b_2 - a_1b_3 + a_2b_0 - a_3b_1)e_4 + (a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0)e_6 \Big) = \\
&0e_0 + 0e_2 + 0e_4 + 0e_6 = 0 \implies oo_1 = o_1o.
\end{aligned}$$

Therefore, $\mathcal{O}_{0246}$ is commutative. This concludes the proof. $\square$

Theorem 6. $\mathcal{O}_{0246}$ is an associative algebra.

Proof. We will show $\mathcal{O}_{0246}$ is an associative algebra with the associator. First, let $o = a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6$, $o_1 = b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6$ and $o_2 = c_0e_0 + c_1e_2 + c_2e_4 + c_3e_6$, e.g. $o, o_1, o_2 \in \mathcal{O}_{0246}$. Then, we have

$$\begin{aligned}
[o, o_1, o_2] &= (oo_1)o_2 - o(o_1o_2) = \\
&\left((a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6)(b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6) \right)(c_0e_0 + c_1e_2 + c_2e_4 + c_3e_6) - \\
&(a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6) \left((b_0e_0 + b_1e_2 + b_2e_4 + b_3e_6)(c_0e_0 + c_1e_2 + c_2e_4 + c_3e_6) \right) = \\
&\left(\underbrace{(a_0b_0 - a_1b_1 - a_2b_2 + a_3b_3)}_{\alpha} e_0 + \underbrace{(a_0b_1 + a_1b_0 - a_2b_3 - a_3b_2)}_{\beta} e_2 + \right. \\
&\left. \underbrace{(a_0b_2 - a_1b_3 + a_2b_0 - a_3b_1)}_{\gamma} e_4 + \underbrace{(a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0)}_{\delta} e_6 \right) (c_0e_0 + c_1e_2 + c_2e_4 + c_3e_6) - \\
&(a_0e_0 + a_1e_2 + a_2e_4 + a_3e_6) \left(\underbrace{(b_0c_0 - b_1c_1 - b_2c_2 + b_3c_3)}_{\hat{\alpha}} e_0 + \underbrace{(b_0c_1 + b_1c_0 - b_2c_3 - b_3c_2)}_{\hat{\beta}} e_2 + \right. \\
&\left. \underbrace{(b_0c_2 - b_1c_3 + b_2c_0 - b_3c_1)}_{\hat{\gamma}} e_4 + \underbrace{(b_0c_3 + b_1c_2 + b_2c_1 + b_3c_0)}_{\hat{\delta}} e_6 \right) = (\alpha c_0 - \beta c_1 - \gamma c_2 + \delta c_3)e_0 + \\
&(\beta c_0 + \alpha c_1 - \delta c_2 - \gamma c_3)e_2 + (\gamma c_0 - \delta c_1 + \alpha c_2 - \beta c_3)e_4 + (\delta c_0 + \gamma c_1 + \beta c_2 + \alpha c_3)e_6 - \\
&\left((a_0\hat{\alpha} - a_1\hat{\beta} - a_2\hat{\gamma} + a_3\hat{\delta})e_0 + (a_0\hat{\beta} + a_1\hat{\alpha} - a_2\hat{\delta} - a_3\hat{\gamma})e_2 + (a_0\hat{\gamma} - a_1\hat{\delta} + a_2\hat{\alpha} - a_3\hat{\beta})e_4 \right. \\
&\left. + (a_0\hat{\delta} + a_1\hat{\gamma} + a_2\hat{\beta} + a_3\hat{\alpha})e_6 \right)
\end{aligned}$$

We can now match the coefficients before the basis vectors, if these are zero then the algebra is associative.

$$\begin{cases} \alpha c_0 - \beta c_1 - \gamma c_2 + \delta c_3 - (a_0 \hat{\alpha} - a_1 \hat{\beta} - a_2 \hat{\gamma} + a_3 \hat{\delta}) = 0? & (i) \\ \beta c_0 + \alpha c_1 - \delta c_2 - \gamma c_3 - (a_0 \hat{\beta} + a_1 \hat{\alpha} - a_2 \hat{\delta} - a_3 \hat{\gamma}) = 0? & (ii) \\ \gamma c_0 - \delta c_1 + \alpha c_2 - \beta c_3 - (a_0 \hat{\gamma} - a_1 \hat{\delta} + a_2 \hat{\alpha} - a_3 \hat{\beta}) = 0? & (iii) \\ \delta c_0 + \gamma c_1 + \beta c_2 + \alpha c_3 - (a_0 \hat{\delta} + a_1 \hat{\gamma} + a_2 \hat{\beta} + a_3 \hat{\alpha}) = 0? & (iv) \end{cases}$$

We show the calculation for (i) first:

$$\begin{aligned} \alpha c_0 - \beta c_1 - \gamma c_2 + \delta c_3 - (a_0 \hat{\alpha} - a_1 \hat{\beta} - a_2 \hat{\gamma} + a_3 \hat{\delta}) &= \underbrace{a_0 b_0 c_0}_{A} - \underbrace{a_1 b_1 c_0}_{B} - \underbrace{a_2 b_2 c_0}_{C} + \underbrace{a_3 b_3 c_0}_{D} \\ &- (\underbrace{a_0 b_1 c_1}_{E} + \underbrace{a_1 b_0 c_1}_{F} - \underbrace{a_2 b_3 c_1}_{G} - \underbrace{a_3 b_2 c_1}_{H}) - (\underbrace{a_0 b_2 c_2}_{I} - \underbrace{a_1 b_3 c_2}_{J} + \underbrace{a_2 b_0 c_2}_{K} - \underbrace{a_3 b_1 c_2}_{L}) \\ &+ \underbrace{a_0 b_3 c_3}_{M} + \underbrace{a_1 b_2 c_3}_{N} + \underbrace{a_2 b_1 c_3}_{O} + \underbrace{a_3 b_0 c_3}_{P} - (\underbrace{a_0 b_0 c_0}_{A} - \underbrace{a_0 b_1 c_1}_{E} - \underbrace{a_0 b_2 c_2}_{I} + \underbrace{a_0 b_3 c_3}_{M} \\ &- (\underbrace{a_1 b_0 c_1}_{F} + \underbrace{a_1 b_1 c_0}_{B} - \underbrace{a_1 b_2 c_3}_{N} - \underbrace{a_1 b_3 c_2}_{J}) - (\underbrace{a_2 b_0 c_2}_{K} - \underbrace{a_2 b_1 c_3}_{O} + \underbrace{a_2 b_2 c_0}_{C} - \underbrace{a_2 b_3 c_1}_{G}) \\ &+ \underbrace{a_3 b_0 c_3}_{P} + \underbrace{a_3 b_1 c_2}_{L} + \underbrace{a_3 b_2 c_1}_{H} + \underbrace{a_3 b_3 c_0}_{D}) = A - A - B - B - C + C + D - D - E + E \\ &- F + F + G - G + H - H - I + I + J - J - K + K + L - L + M - M + N - N + \\ &O - O + P - P = 0. \end{aligned}$$

The calculation for (ii):

$$\begin{aligned} \beta c_0 + \alpha c_1 - \delta c_2 - \gamma c_3 - (a_0 \hat{\beta} + a_1 \hat{\alpha} - a_2 \hat{\delta} - a_3 \hat{\gamma}) &= \underbrace{a_0 b_1 c_0}_{A} + \underbrace{a_1 b_0 c_0}_{B} - \underbrace{a_2 b_3 c_0}_{C} - \underbrace{a_3 b_2 c_0}_{D} \\ &+ \underbrace{a_0 b_0 c_1}_{E} - \underbrace{a_1 b_1 c_1}_{F} - \underbrace{a_2 b_2 c_1}_{G} + \underbrace{a_3 b_3 c_1}_{H} - (\underbrace{a_0 b_3 c_2}_{I} + \underbrace{a_1 b_2 c_2}_{J} + \underbrace{a_2 b_1 c_2}_{K} + \underbrace{a_3 b_0 c_2}_{L}) \\ &- (\underbrace{a_0 b_2 c_3}_{M} - \underbrace{a_1 b_3 c_3}_{N} + \underbrace{a_2 b_0 c_3}_{O} - \underbrace{a_3 b_1 c_3}_{P}) - (\underbrace{a_0 b_0 c_1}_{E} + \underbrace{a_0 b_1 c_0}_{A} - \underbrace{a_0 b_2 c_3}_{M} - \underbrace{a_0 b_3 c_2}_{I} \\ &+ \underbrace{a_1 b_0 c_0}_{B} - \underbrace{a_1 b_1 c_1}_{F} - \underbrace{a_1 b_2 c_2}_{J} + \underbrace{a_1 b_3 c_3}_{N} - (\underbrace{a_2 b_0 c_3}_{O} + \underbrace{a_2 b_1 c_2}_{K} + \underbrace{a_2 b_2 c_1}_{G} + \underbrace{a_2 b_3 c_0}_{C}) \\ &- (\underbrace{a_3 b_0 c_2}_{L} - \underbrace{a_3 b_1 c_3}_{P} + \underbrace{a_3 b_2 c_0}_{D} - \underbrace{a_3 b_3 c_1}_{H})) = A - A + B - B - C + C - D + D + E - E \\ &- F + F - G + G + H - H - I + I - J + J - K + K - L + L - M + M + N - N \\ &- O + O + P - P = 0. \end{aligned}$$

The calculation for (iii):

$$\begin{aligned}
& \gamma c_0 - \delta c_1 + \alpha c_2 - \beta c_3 - (a_0 \hat{\gamma} - a_1 \hat{\delta} + a_2 \hat{\alpha} - a_3 \hat{\beta}) = \underbrace{a_0 b_2 c_0}_{A} - \underbrace{a_1 b_3 c_0}_{B} + \underbrace{a_2 b_0 c_0}_{C} - \underbrace{a_3 b_1 c_0}_{D} \\
& - (\underbrace{a_0 b_3 c_1}_{E} + \underbrace{a_1 b_2 c_1}_{F} + \underbrace{a_2 b_1 c_1}_{G} + \underbrace{a_3 b_0 c_1}_{H}) + \underbrace{a_0 b_0 c_2}_{I} - \underbrace{a_1 b_1 c_2}_{J} - \underbrace{a_2 b_2 c_2}_{K} + \underbrace{a_3 b_3 c_2}_{L} \\
& - (\underbrace{a_0 b_1 c_3}_{M} + \underbrace{a_1 b_0 c_3}_{N} - \underbrace{a_2 b_3 c_3}_{O} - \underbrace{a_3 b_2 c_3}_{P}) - \left(\underbrace{a_0 b_0 c_2}_{I} - \underbrace{a_0 b_1 c_3}_{M} + \underbrace{a_0 b_2 c_0}_{A} - \underbrace{a_0 b_3 c_1}_{E} \right. \\
& - (\underbrace{a_1 b_0 c_3}_{N} + \underbrace{a_1 b_1 c_2}_{J} + \underbrace{a_1 b_2 c_1}_{F} + \underbrace{a_1 b_3 c_0}_{B}) + \underbrace{a_2 b_0 c_0}_{C} - \underbrace{a_2 b_1 c_1}_{G} - \underbrace{a_2 b_2 c_2}_{K} + \underbrace{a_2 b_3 c_3}_{O} \\
& \left. - (\underbrace{a_3 b_0 c_1}_{H} + \underbrace{a_3 b_1 c_0}_{D} - \underbrace{a_3 b_2 c_3}_{P} - \underbrace{a_3 b_3 c_2}_{L}) \right) = A - A - B + B + C - C - D + D - E + E \\
& - F + F - G + G - H + H + I - I - J + J - K + K + L - L - M + M - N + N \\
& + O - O + P - P = 0.
\end{aligned}$$

The calculation for (iv):

$$\begin{aligned}
& \delta c_0 + \gamma c_1 + \beta c_2 + \alpha c_3 - (a_0 \hat{\delta} + a_1 \hat{\gamma} + a_2 \hat{\beta} + a_3 \hat{\alpha}) = \underbrace{a_0 b_3 c_0}_{A} + \underbrace{a_1 b_2 c_0}_{B} + \underbrace{a_2 b_1 c_0}_{C} + \underbrace{a_3 b_0 c_0}_{D} \\
& + \underbrace{a_0 b_2 c_1}_{E} - \underbrace{a_1 b_3 c_1}_{F} + \underbrace{a_2 b_0 c_1}_{G} - \underbrace{a_3 b_1 c_1}_{H} + \underbrace{a_0 b_1 c_2}_{I} + \underbrace{a_1 b_0 c_2}_{J} - \underbrace{a_2 b_3 c_2}_{K} - \underbrace{a_3 b_2 c_2}_{L} \\
& + \underbrace{a_0 b_0 c_3}_{M} - \underbrace{a_1 b_1 c_3}_{N} - \underbrace{a_2 b_2 c_3}_{O} + \underbrace{a_3 b_3 c_3}_{P}) - \left(\underbrace{a_0 b_0 c_3}_{M} + \underbrace{a_0 b_1 c_2}_{I} + \underbrace{a_0 b_2 c_1}_{E} + \underbrace{a_0 b_3 c_0}_{A} \right. \\
& + \underbrace{a_1 b_0 c_2}_{J} - \underbrace{a_1 b_1 c_3}_{N} + \underbrace{a_1 b_2 c_0}_{B} - \underbrace{a_1 b_3 c_1}_{F} + \underbrace{a_2 b_0 c_1}_{G} + \underbrace{a_2 b_1 c_0}_{C} - \underbrace{a_2 b_2 c_3}_{O} - \underbrace{a_2 b_3 c_2}_{K} \\
& \left. + \underbrace{a_3 b_0 c_0}_{D} - \underbrace{a_3 b_1 c_1}_{H} - \underbrace{a_3 b_2 c_2}_{L} + \underbrace{a_3 b_3 c_3}_{P} \right) = A - A + B - B + C - C + D - D + E - E \\
& - F + F + G - G - H + H + I - I + J - J - K + K - L + L + M - M - N + N \\
& - O + O + P - P = 0.
\end{aligned}$$

We have shown that the associator equals zero for each basis vector and herefore, it follows that $[o, o_1, o_2] = (oo_1)o_2 - o(o_1o_2) = 0 \iff (oo_1)o_2 = o(o_1o_2)$. Hence O_{0246} is an associative algebra. This completes the proof. $\square$

Proposition 27. Let $G = \{\pm e_0, \pm e_2, \pm e_4, \pm e_6\}$. This is the basis for the vector space O_{0246} over $\mathbb{R}$ with their additive inverses. Then G is an abelian group.

Proof. Since O_{0246} is a commutative and associative algebra, the proof follows. The identity is e_0 and $\#G = 8$. $\square$

We present the multiplication table, again. The red indicates all basis elements that commute with each other.

	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	$-e_0$	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	$-e_0$	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	e_6	e_7	$-e_0$	e_1	$-e_2$	$-e_3$
e_5	e_5	e_4	e_7	$-e_6$	$-e_1$	$-e_0$	e_3	$-e_2$
e_6	e_6	e_7	$-e_4$	e_5	$-e_2$	e_3	e_0	e_1
e_7	e_7	$-e_6$	$-e_5$	$-e_4$	$-e_3$	$-e_2$	$-e_1$	e_0

Table 4.5: Commutative multiplication table for $\mathcal{O}$ as seen as a vector space over $\mathbb{R}$.

From the red cells in Table 4.5, we see that $e_2e_3 \neq e_3e_2$, $e_4e_5 \neq e_5e_4$ and $e_6e_7 \neq e_7e_6$. We propose that a set of basis vectors that commutes (the red) is a commutative algebra if they are chosen such that if e_2 is in the basis, then e_3 is not, if e_4 is in the basis, then e_5 is not and also if e_6 is in the basis e_7 is not.

We have already shown that a subalgebra of dimension 4 exists which is commutative- the algebra $\mathcal{O}_{0246}$. This algebra is also associative. A different set of basis vectors that is closed under multiplication is $\mathcal{B}_{\mathcal{O}_{0257}} = \{e_0, e_2, e_5, e_7\}$. We state that now.

Conjecture 2. *The set of vectors $\mathcal{B}_{\mathcal{O}_{0257}} = \{e_0, e_2, e_5, e_7\}$ span a 4-dimensional vector space over $\mathbb{R}$ and form a 4-dimensional commutative and associative subalgebra of $\mathcal{O}$. We call the algebra $\mathcal{O}_{0257}$.*

We observe that there are four different set of basis vectors of dimension 4 that commute and are closed. In addition to $\mathcal{O}_{0246}$ and $\mathcal{O}_{0257}$, these are $\mathcal{O}_{0347}$ and $\mathcal{O}_{0356}$. We believe all of these algebras are commutative and associative.

Conjecture 3. *The set of vectors $\mathcal{B}_{\mathcal{O}_{0347}}$ and $\mathcal{B}_{\mathcal{O}_{0356}}$ span 4-dimensional vector spaces over $\mathbb{R}$ and form 4-dimensional subalgebras of $\mathcal{O}$. Both algebras are commutative and associative.*

Furthermore, collect these 4 subalgebras in a set $A = \{\mathcal{O}_{0246}, \mathcal{O}_{0257}, \mathcal{O}_{0347}, \mathcal{O}_{0356}\}$.

Conjecture 4. *Each algebra in A have left and right zero divisors of degree 2.*

We already know this is true for the algebra $\mathcal{O}_{0246}$ and it seems logical that it holds for the others as well. The basis vectors for each algebra in A might also form an abelian group.

Conjecture 5. *Let $\mathcal{B}_{\hat{\mathcal{O}}}$ be the basis for any algebra $B \in A$. By ${}^{-}\mathcal{B}_{\hat{\mathcal{O}}}$ we mean the additive inverse for each basis element in $\mathcal{B}_{\hat{\mathcal{O}}}$. Then, $G = \mathcal{B}_{\hat{\mathcal{O}}} \cup {}^{-}\mathcal{B}_{\hat{\mathcal{O}}}$ form an abelian group with cardinality $\#G = 8$.*

Furthermore, we have 8 subalgebras of dimension 2, these have basis $\mathcal{B} = \{e_0, e_i\}$ where $i = 1, 2, \dots, 7$. Call such a 2-dimensional algebra O_i where i is the subscript for the other basis vector. Then we have.

Proposition 28. *The algebras O_i for $i = 1, 2, 3, 4, 5$ are isomorphic to $\mathbb{C}$.*

Proof. Let $o_i = a_0 e_0 + a_1 e_i$. Then, the mapping $(a_0, a_1) \mapsto a + bi$ makes these algebras isomorphic. To prove that this mapping is an isomorphism is the same as in Proposition 7 in section 3.1.1. $\square$

However, only $O_1 \cong^* \mathbb{C}$ with ordinary complex conjugation on $\mathbb{C}$. Furthermore, we have that the algebras O_i for $i = 6, 7$ are isomorphic to $\mathbb{C}'$ by similar arguments. However, these algebras are not $*$ -isomorphic to the split-complexes. Lastly, we have some propositions if we were to utilize the Cayley–Dickson construction again, meaning $\text{Cay}(O, -1)$.

Conjecture 6. *The amount of elements that changes in the multiplication table for $\text{Cay}(O, -1)$ is 2^5 compared to the multiplication for $\text{Cay}(\mathbb{O}, -1)$. And in general for doubling step n , we have 2^{n+2} changes compared to the standard doubling procedure, if we start with $\mathbb{C}$.*

Conjecture 7. *The amount of subalgebras of dimension 4 in $\text{Cay}(O, -1)$ is 2^3 and the amount of subalgebras of dimension 8 in $\text{Cay}(O, -1)$ is 2^2 . All these algebras are commutative and associative.*

4.3.3 $\text{Cay}_+(\mathbb{C}, 1)$

We start with $\mathbb{C}$ with the ordinary complex conjugation. Thereafter, we construct the algebra $\text{Cay}_+(\mathbb{C}, 1) = \mathcal{H}'$. The vector space is $\mathbb{C} \oplus \mathbb{C}$ or $\bigoplus_{i=1}^4 \mathbb{R}$. Therefore a basis over $\mathbb{C}$ would be $\mathcal{B}_C = \{1, j\}$ and basis over $\mathbb{R}$ would be $\mathcal{B}_R = \{1, i, j, k\}$. We show the multiplication tables first.

	$(1, 0)$	$(i, 0)$	$(0, 1)$	$(0, i)$
$(1, 0)$	$(1, 0)$	$(i, 0)$	$(0, 1)$	$(0, i)$
$(i, 0)$	$(i, 0)$	$-(1, 0)$	$(0, i)$	$-(0, 1)$
$(0, 1)$	$(0, 1)$	$-(0, i)$	$(1, 0)$	$-(i, 0)$
$(0, i)$	$(0, i)$	$(0, 1)$	$(i, 0)$	$(1, 0)$

Table 4.6: Multiplication table for $\text{Cay}(\mathbb{C}, 1)$ as seen as a vector space over $\mathbb{R}$.

	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	1	$-i$
k	k	j	i	1

Table 4.7: Multiplication table for $\mathcal{H}'$ as seen as a vector space over $\mathbb{R}$.

We observe that the multiplication table 4.7 is different than for the algebra $Cay_+(\mathbb{C}, -1)$. Where the green indicates the difference from the multiplication table for $Cay(\mathbb{C}, -1)$. In fact, this multiplication is the same as for the split quaternions and therefore we have $Cay(\mathbb{C}, 1) \cong \mathbb{H}'$. From the discussion about split-quaternions, we said they were not commutative but associative, without proof. We show that $Cay(\mathbb{C}, 1)$ is not commutative but associative now, which indirectly also proves the statements for the split-quaternions.

Proposition 29. *$Cay(\mathbb{C}, 1)$ is not a commutative algebra.*

Proof. Let $(i, 0), (0, 1) \in Cay(\mathbb{C}, 1)$. Then, by the commutator, we have

$$\begin{aligned} [(i, 0), (0, 1)] &= (i, 0)(0, 1) - (0, 1)(i, 0) = \\ &= (i \cdot 0 + 1 \cdot 0, 1 \cdot i + 0) - (0 \cdot i + 0 \cdot 1, 0 + 1 \cdot i^*) = \\ &= (0, i) - (0, -i) = (0, 2i) \neq (0, 0). \end{aligned}$$

Since commutativity does not hold for the elements $(i, 0)$ and $(0, 1)$, $Cay(\mathbb{C}, 1)$ is not commutative. This completes the proof. $\square$

Proposition 30. *$Cay(\mathbb{C}, 1)$ is an associative algebra.*

Proof. Let $(a, b), (c, d), (e, f) \in Cay(\mathbb{C}, 1)$ where $a, b, \dots, f \in \mathbb{C}$. Then, by using the associator, we have

$$\begin{aligned} [(a, b), (c, d), (e, f)] &= \left((a, b)(c, d) \right) (e, f) - (a, b) \left((c, d)(e, f) \right) = \\ &= (ac + d^*b, da + bc^*)(e, f) - (a, b)(ce + f^*d, fc + de^*) = \\ &= ((ac + d^*b)e + f^*(da + bc^*), f(ac + d^*b) + (da + bc^*)e^*) \\ &\quad - (a(ce + f^*d) + (fc + de^*)^*b, (fc + de^*)a + b(ce + f^*d)^*) = \\ &= (ace + d^*be + f^*da + f^*bc^*, fac + fd^*b + dae^* + bc^*e^*) \\ &\quad - (ace + af^*d + (c^*f^* + ed^*)b, fca + de^*a + b(e^*c^* + d^*f)) = \\ &= (ace + d^*be + f^*da + f^*bc^*, fac + fd^*b + dae^* + bc^*e^*) \\ &\quad - (ace + af^*d + c^*f^*d + ed^*b, fca + de^*a + be^*c^* + bd^*f) = (0, 0) \implies \\ &= \left((a, b)(c, d) \right) (e, f) = (a, b) \left((c, d)(e, f) \right). \end{aligned}$$

We have utilized that the complex numbers are both commutative and associative. This concludes the proof. $\square$

We should note that the basis vectors under conjugation gives:

$$\begin{cases} 1^* = 1 \in \mathbb{H}' \text{ is represented by } (1, 0)^* = (1^*, 0) = (1, 0) \in Cay_+(\mathbb{C}, 1), \\ i^* = -i \in \mathbb{H}' \text{ is represented by } (i, 0)^* = (i^*, 0) = (-i, 0) \in Cay_+(\mathbb{C}, 1), \\ j^* = j \in \mathbb{H}' \text{ is represented by } (0, 1)^* = (0, 1) = (0, 1) \in Cay_+(\mathbb{C}, 1), \\ k^* = k \in \mathbb{H}' \text{ is represented by } (0, i)^* = (0, i) = (0, i) \in Cay_+(\mathbb{C}, 1). \end{cases}$$

We have that an element in $h \in Cay_+(\mathbb{C}, 1)$ can be written, where $\lambda_i \in \mathbb{R} \ i = 1, 2, 3, 4$, as: $h = \lambda_1 + \lambda_2i + \lambda_3j + \lambda_4k$ and where conjugation is given by $h^* = \lambda_1 - \lambda_2i + \lambda_3j + \lambda_4k$. This conjugation is different than for ordinary split-quaternions and hence $Cay_+(\mathbb{C}, 1) \not\cong \mathbb{H}'$.

4.3.4 $\text{Cay}_+(\mathcal{H}', 1)$

We continue the doubling of algebras to obtain the vector space $\mathcal{H}' \oplus \mathcal{H}'$ as seen over $\mathcal{H}'$ or the vector space $\bigoplus_{i=1}^8 \mathbb{R}$ over $\mathbb{R}$. This is an 8-dimensional vector space over $\mathbb{R}$. Hence a basis is $\mathcal{B} = \{e_0, e_1, \dots, e_7\}$. A basis for $\mathcal{H}' \oplus \mathcal{H}'$ as seen over $\mathcal{H}'$ would be $\mathcal{B}_{\mathcal{H}'} = \{1, l\}$ where each element in $\text{Cay}_+(\mathcal{H}', 1) = \mathcal{O}'$ would be written as $o = a_1 + a_2 l$ and where $a_1, a_2 \in \text{Cay}(\mathcal{H}', 1)$. Furthermore, we observe that the basis vectors $e_i = (a_1, a_2)$ where $a_1, a_2 \in \mathcal{H}'$ and $i = 0, 1, \dots, 7$. We show the different representations, now.

$e_0 \in \mathcal{O}'$ is represented by $(1, 0) \in \text{Cay}(\mathcal{H}', 1)$ $e_1 \in \mathcal{O}'$ is represented by $(i, 0) \in \text{Cay}(\mathcal{H}', 1)$
 $e_2 \in \mathcal{O}'$ is represented by $(j, 0) \in \text{Cay}(\mathcal{H}', 1)$ $e_3 \in \mathcal{O}'$ is represented by $(k, 0) \in \text{Cay}(\mathcal{H}', 1)$
 $e_4 \in \mathcal{O}'$ is represented by $(0, 1) \in \text{Cay}(\mathcal{H}', 1)$ $e_5 \in \mathcal{O}'$ is represented by $(0, i) \in \text{Cay}(\mathcal{H}', 1)$
 $e_6 \in \mathcal{O}'$ is represented by $(0, j) \in \text{Cay}(\mathcal{H}', 1)$ $e_7 \in \mathcal{O}'$ is represented by $(0, k) \in \text{Cay}(\mathcal{H}', 1)$

We present the multiplication tables for the algebra $\text{Cay}(\mathcal{H}', 1)$, now.

	$(1, 0)$	$(i, 0)$	$(j, 0)$	$(k, 0)$	$(0, 1)$	$(0, i)$	$(0, j)$	$(0, k)$
$(1, 0)$	$(1, 0)$	$(i, 0)$	$(j, 0)$	$(k, 0)$	$(0, 1)$	$(0, i)$	$(0, j)$	$(0, k)$
$(i, 0)$	$(i, 0)$	$-(1, 0)$	$(k, 0)$	$-(j, 0)$	$(0, i)$	$-(0, 1)$	$-(0, k)$	$(0, j)$
$(j, 0)$	$(j, 0)$	$-(k, 0)$	$(1, 0)$	$-(i, 0)$	$(0, j)$	$(0, k)$	$(0, 1)$	$(0, i)$
$(k, 0)$	$(k, 0)$	$(j, 0)$	$(i, 0)$	$(1, 0)$	$(0, k)$	$-(0, j)$	$-(0, i)$	$(0, 1)$
$(0, 1)$	$(0, 1)$	$-(0, i)$	$(0, j)$	$(0, k)$	$(1, 0)$	$-(i, 0)$	$(j, 0)$	$(k, 0)$
$(0, i)$	$(0, i)$	$(0, 1)$	$(0, k)$	$-(0, j)$	$(i, 0)$	$(1, 0)$	$-(k, 0)$	$(j, 0)$
$(0, j)$	$(0, j)$	$(0, k)$	$(0, 1)$	$-(0, i)$	$(j, 0)$	$-(k, 0)$	$(1, 0)$	$(i, 0)$
$(0, k)$	$(0, k)$	$-(0, j)$	$(0, i)$	$(0, 1)$	$(k, 0)$	$(j, 0)$	$-(i, 0)$	$(1, 0)$

Table 4.8: Multiplication table for $\text{Cay}(\mathcal{H}', 1)$ as seen as a vector space over $\mathbb{R}$.

	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	e_0	$-e_1$	e_6	e_7	e_4	e_5
e_3	e_3	e_2	e_1	e_0	e_7	$-e_6$	$-e_5$	e_4
e_4	e_4	$-e_5$	e_6	e_7	e_0	$-e_1$	e_2	e_3
e_5	e_5	e_4	e_7	$-e_6$	e_1	e_0	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	e_2	$-e_3$	e_0	e_1
e_7	e_7	$-e_6$	e_5	e_4	e_3	e_2	$-e_1$	e_0

Table 4.9: Multiplication table for $\mathcal{O}'$, as seen as a vector space over $\mathbb{R}$.

The green cells in Table 4.9 indicates a change compared to the basis vector multiplication of the split-octonions in Table 3.5. Furthermore, each element in $\mathcal{O}'$ can be written as

$o = \sum_{i=0}^7 a_i e_i$ where $a_i \in \mathbb{R} \ i = 1, 2, \dots, 7$. It follows from the involution on $\mathcal{H}'$ that

$$o^* = a_0 e_0 - a_1 e_1 + \sum_{i=2}^7 a_i e_i.$$

We know that the only basis element that flips sign under conjugation is e_1 , e.g. $e_1^* = -e_1$. Otherwise we have $e_i^* = e_i$ for $i = 0, 2, 3, \dots, 7$. As with the split-octonions, we suspect that $\text{Cay}(\mathcal{H}', 1)$ is not commutative, which we show first.

Proposition 31. *$\text{Cay}(\mathcal{H}', 1)$ is not a commutative algebra.*

Proof. Let $e_2, e_3 \in \text{Cay}(\mathcal{H}', 1)$. Then, by the commutator, we have

$$[e_2, e_3] = e_2 e_3 - e_3 e_2 = -e_1 - e_1 = -2e_1 \neq 0.$$

Since the commutator is nonzero, $\text{Cay}(\mathcal{H}', 1)$ is not commutative. This completes the proof. $\square$

Proposition 32. *$\text{Cay}(\mathcal{H}', 1)$ is not an associative algebra.*

Proof. Let $e_2, e_3, e_4 \in \text{Cay}(\mathcal{H}', 1)$. Then, by using the associator, we have

$$[e_2, e_3, e_4] = (e_2 e_3) e_4 - e_2 (e_3 e_4) = -e_1 e_4 - e_2 e_7 = -e_5 - e_5 = -2e_5 \neq 0.$$

Since the associator is nonzero, $\text{Cay}(\mathcal{H}', 1)$ is not associative. This completes the proof. $\square$

	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	e_0	$-e_1$	e_6	e_7	e_4	e_5
e_3	e_3	e_2	e_1	e_0	e_7	$-e_6$	$-e_5$	e_4
e_4	e_4	$-e_5$	e_6	e_7	e_0	$-e_1$	e_2	e_3
e_5	e_5	e_4	e_7	$-e_6$	e_1	e_0	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	e_2	$-e_3$	e_0	e_1
e_7	e_7	$-e_6$	e_5	e_4	e_3	e_2	$-e_1$	e_0

Table 4.10: Commutative multiplication table for $\mathcal{O}'$, as seen as a vector space over $\mathbb{R}$.

Similarly to the algebra $\mathcal{O}$, we have an identical form of red in the table. From the red cells in Table 4.10, we see that $e_2 e_3 \neq e_3 e_2$, $e_4 e_5 \neq e_5 e_4$ and $e_6 e_7 \neq e_7 e_6$. We propose that a set of basis vectors that commutes (the red) is a commutative algebra if they are chosen such that if e_2 is in the basis, then e_3 is not, if e_4 is in the basis, then e_5 is not and also if e_6 is in the basis e_7 is not.

We observe that there are four different set of basis vectors of dimension 4 that commute and are closed. These are $\mathcal{O}'_{0246}$, $\mathcal{O}'_{0257}$, $\mathcal{O}'_{0347}$ and $\mathcal{O}'_{0356}$. Furthermore, collect these 4 different sets of basis vectors in an union of sets $A' = \{\mathcal{O}'_{0246}, \mathcal{O}'_{0257}, \mathcal{O}'_{0347}\} \cup \{\mathcal{O}'_{0356}\} = A'_1 \cup A'_2$. Each set of basis vectors in A' form a 4-dimensional vector space over $\mathbb{R}$. Therefore, we propose similar conjectures as for the subalgebras of $\text{Cay}(\mathcal{H}, -1)$.

Conjecture 8. *Each vector space in A' form a 4-dimensional subalgebra of O' and is a commutative and associative algebra over $\mathbb{R}$.*

Conjecture 9. *Each algebra in A' have left and right zero divisors of degree 2.*

Conjecture 10. *Let $\mathcal{B}_{\hat{O}}$, be any basis for any algebra $B' \in A'_1$. Then $G = \mathcal{B}_{\hat{O}}$, form an abelian group with cardinality $\#G = 4$.*

Conjecture 11. *Let $\mathcal{B}_{\hat{O}}$, be the basis for the algebra $B' \in A'_2$. By ${}^{-}\mathcal{B}_{\hat{O}}$, we mean the additive inverses for the basis elements. Then $G = \mathcal{B}_{\hat{O}} \cup {}^{-}\mathcal{B}_{\hat{O}}$, form an abelian group with cardinality $\#G = 8$.*

Chapter 5

Discussion and Final Remarks

By utilizing the Cayley–Dickson construction with the standard involution $(a, b)^* = (a^*, -b)$ on the algebra $\mathbb{R}$ over $\mathbb{R}$ with parameter $\mu = -1$, we have shown that we obtain algebras of twice the dimension of the previous ones. Furthermore, we have shown that these algebras are $*$ -algebra isomorphic to the complex numbers, the quaternions and the octonions. A natural norm with the use of multiplication of an element with its conjugation has likewise been shown. We know by Wedderburn, that these four algebras are the only real absolute valued algebras. The real absolute valued algebras have all been well studied. We also commented on the split algebras obtained with the choice of parameter $\mu = 1$.

Thereafter we showed that $(a, b)^* = (a^*, b)$ is also an involution on $\text{Cay}(A, \mu)$. With that knowledge, we started doubling the algebras, starting with the algebra $\mathbb{R}$ over itself. After some research, it was found that after starting with $\mathbb{R}$ and the identity involution in the said algebra that the subsequently algebras were all commutative and associative. This ought to be trivial upon reflection since this new involution is the identity mapping almost on the bigger algebra obtained through the Cayley–Dickson construction. Therefore, we started the process of creating new algebras with the involution $(a, b)^* = (a^*, b)$ and by starting with the algebra $\mathbb{C}$ over $\mathbb{R}$ with the involution on $\mathbb{C}$ given by the ordinary complex conjugation.

This process was first performed with $\mu = -1$. By constructing $\text{Cay}_+(\mathbb{C}, -1) = \mathcal{H}$ we obtained a 4-dimensional algebra which was isomorphic to the quaternions, although not $*$ -isomorphic. Therefore, all properties that $\mathbb{H}$ has which does not involve the conjugation is shared with $\mathcal{H}$. This is not too surprising, as we have only done one step and the new involution, $(a, b)^* = (a^*, b)$ has not yet left its mark on our new algebras to a great extent. Therefore, the more interesting algebra came by doubling again, e.g. by constructing $\text{Cay}_+(\mathcal{H}, -1) = \mathcal{O}$. We found that $\mathcal{O} \not\cong \mathbb{O}$, which means that it must contain zero divisors. By examining the algebra, we found that there are zero divisors on the form $o = a_o e_i + e_j$ for some basis vectors $e_i \neq e_j$ and $i, j = 0, 1, 2, \dots, 7$. The fact that there are zero divisors which commute is very interesting. Furthermore, if there are zero divisors consisting of more basis vectors than 2, then the amount of basis vectors must be a power of 2. We also showed that there are left and right zero divisors of degree 4- meaning with 4 basis vectors. Although we are more skeptical of zero divisors of degree 8- consisting of all basis vectors. However, if they exist they do not commute and there-

fore there might only be left or right zero divisors of degree 8. More research on this matter to finally conclude whether the statement is false or true would be significant. The algebra $\mathcal{O}$ also have at least one subalgebra of dimension 4, which is both commutative and associative. This algebra is different than the quaternions and hence that algebra is not isomorphic to the quaternions as the quaternions are not commutative. However, we believe that there are four different 4-dimensional subalgebras of $\mathcal{O}$. More research into this matter and also to look for isomorphisms between these algebras and other known algebras should be conducted for a greater understanding of the Cayley–Dickson construction with the other involution. We also believe that the pattern continues if we were to double the algebra $\mathcal{O}$, meaning we ought to increase the amount of 4-dimensional algebras of some power, presumably 2, but also generate subalgebras of dimension 8. The subalgebras of dimension 8 might or might not be commutative and/or associative. Examining the algebra $\text{Cay}(\mathcal{O}, -1)$ with involution $(a, b)^* = (a^*, b)$ is therefore of great interest. Furthermore, if there are no zero divisors of degree 8 in $\mathcal{O}$, there ought to be such zero divisors in $\text{Cay}(\mathcal{O}, -1)$ as an 8-dimensional algebra in the basis given by $\text{Cay}(\mathcal{O}, -1)$ would be a subalgebra as $\text{Cay}(\mathcal{O}, -1)$ would be 16-dimensional. If there are no zero divisors of degree 8 in $\mathcal{O}$, we believe there could not exist zero divisors of degree 16 in $\text{Cay}(\mathcal{O}, -1)$. Research into this question in order to propose and prove the amount of zero divisors, of which degrees and the amount of subalgebras of different dimensions in the general process by starting with $\mathbb{C}$ and involution $(a, b)^* = (a^*, b)$ would be beneficial.

Subsequently, we redid the process but with $\mu = 1$. Not surprisingly, we obtained an algebra $\text{Cay}(\mathbb{C}, 1) = \mathcal{H}'$, which was isomorphic to the split-quaternions. Although not $*$ -isomorphic. As with $\mu = -1$, nothing else happened in the first step. However, after doubling again to obtain $\text{Cay}(\mathcal{H}', 1) = \mathcal{O}'$, more interesting things happened. Exactly the same number of elements as for $\mathcal{O}$ in the multiplication table for $\mathcal{O}'$, compared to the split-octonions, changed. $\mathcal{O}'$ also had a similar symmetry to $\mathcal{O}$ of the basis vectors that commuted. Therefore, we ought to believe that there are four 4-dimensional subalgebras of $\mathcal{O}'$, all of which we postulate are associative and commutative. The interesting part is that there are exactly the same basis vectors that commute in $\mathcal{O}'$ and $\mathcal{O}$. Research on whether there are 4 subalgebras of dimension 4 for both $\mathcal{O}'$ and $\mathcal{O}$ and if they are isomorphic would be interesting to investigate. Furthermore, if there are any zero divisors, left and/or right, and for which degrees should also be looked into. The same patterns postulated for $\text{Cay}(\mathcal{O}, -1)$ should also be studied if we double $\mathcal{O}'$ by constructing $\text{Cay}(\mathcal{O}', 1)$. We believe that the patterns should be the same for both choices of parameters. But what about their subalgebras of different dimensions? The answers to whether the subalgebras of $\text{Cay}(\mathcal{O}', 1)$ and $\text{Cay}(\mathcal{O}, -1)$ are isomorphic, and, more generally in each subsequent step and similar questions, will be of great interest.

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